Novel imaging techniques for intraoperative margin assessment in surgical oncology: a systematic review.

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Supplementary 1: Methods

1. Search strategy

PubMed

Part 1: Margins of excision
("Margins of excision"[Mesh]
OR
Excision margin [tiab] OR Excision margins [tiab] OR Lesion margin [tiab] OR Lesion margins [tiab] OR Margin assessment [tiab] OR Margin status [tiab] OR Resection margin [tiab] OR Resection margins [tiab] OR Safe margin[tiab] OR Safe margins [tiab] OR Specimen margin[tiab] OR Specimen margins [tiab] OR Surgical margin [tiab] OR Surgical margins [tiab] OR Tumor-free margin [tiab] OR Tumor-free margins [tiab] OR Tumour-free margin [tiab] OR Tumour-free margins [tiab]) AND ("2013/01/01"[PDAT] : "2018/12/31"[PDAT])

Part 2: Diagnostic Imaging
("Diagnostic imaging"[Mesh] OR "diagnostic imaging" [Subheading]
OR
CT [tiab] OR echography [tiab] OR fluorescence [tiab] OR fluoroscopy [tiab] OR image [tiab] OR images [tiab] OR imaging [tiab] OR microscopy [tiab] OR MR [tiab] OR MRI [tiab] OR NIR [tiab] OR OCT [tiab] OR photography [tiab] OR radiography [tiab] OR radioisotope scanning [tiab] OR roentgenography [tiab] OR scintigraphy [tiab] OR scintiphotography [tiab] OR spectroscopy [tiab] OR stroboscopy [tiab] OR thermography [tiab] OR tomography [tiab] OR X-ray [tiab] OR X-rays [tiab] OR ultrasonography [tiab] OR ultrasound [tiab]) AND ("2013/01/01"[PDAT] : "2018/12/31"[PDAT])

Part 3: Histology
(Histology [Mesh] OR Pathology [Mesh] OR Pathology [Subheading]
OR
Histologic [tiab] OR Histological [tiab] OR Histologically [tiab] OR Histology [tiab] OR Histopathologic [tiab] OR Histopathological [tiab] OR Histopathologically [tiab] OR Histopathology [tiab] OR Pathologic [tiab] OR Pathological [tiab] OR pathologically [tiab] OR Pathology [tiab]) AND ("2013/01/01"[PDAT] : "2018/12/31"[PDAT])

Embase

Part 1: Margins of excision
exp surgical margin/
OR
(Excision margin OR Excision margins OR Lesion margin OR Lesion margins OR Margin assessment OR Margin status OR Resection margin OR Resection margins OR Safe margin OR Safe margins OR Specimen margin OR Specimen margins OR Surgical margin OR Surgical margins OR Tumor-free margin OR Tumor-free margins OR Tumour-free margin OR Tumour-free margins).ti,ab,kw.

Part 2: Diagnostic Imaging
exp echography/ OR exp radiodiagnosis/ OR exp spectroscopy/
OR
(CT OR Echography OR fluorescence OR fluoroscopy OR image OR images OR imaging OR microscopy OR MR OR MRI OR NIR OR OCT OR photography OR radiography OR radioisotope scanning OR roentgenography OR scintigraphy OR scintiphotography OR spectroscopy OR stroboscopy OR thermography OR tomography OR ultrasonography OR ultrasound OR X-ray OR X-rays).ti,ab,kw.

Part 3: Histology
exp Histology/ OR exp Histopathology/ OR exp Pathology/
OR
(Histologic OR Histological OR Histologically OR Histology OR Histopathologic OR Histopathological OR Histopathologically OR Histopathology OR Pathologic OR Pathological OR pathologically OR Pathology).ti,ab,kw.

Web of Science

Part 1: Margins of excision
TS = ("Excision margin" OR "Excision margins" OR "Lesion margin" OR "Lesion margins" OR "Margin assessment" OR "Margin status" OR "Resection margin" OR "Resection margins" OR "Safe margin" OR "Safe margins" OR "Specimen margin" OR "Specimen margins" OR "Surgical margin" OR "Surgical margins" OR "Tumor-free margin" OR "Tumor-free margins" OR "Tumour-free margin" OR "Tumour-free margins")

Part 2: Diagnostic imaging
TS = (CT OR echography OR fluorescence OR fluoroscopy OR image OR images OR imaging OR microscopy OR MR OR MRI OR NIR OR OCT OR photography OR radiography OR radioisotope scanning OR roentgenography OR scintigraphy OR scintiphotography OR spectroscopy OR stroboscopy OR thermography OR tomography OR ultrasonography OR ultrasound OR X-ray OR X-rays)

Part 3: Histology

TS = (Histologic OR Histological OR Histologically OR Histology OR Histopathologic OR Histopathological OR Histopathologically OR Histopathology OR Pathologic OR Pathological OR pathologically OR Pathology)

The Cochrane Library

Part 1: Margins of excision
[mh “Margins of excision”]
OR
(“Excision margin” OR “Excision margins” OR “Lesion margin” OR “Lesion margins” OR “Margin assessment” OR “Margin status” OR “Resection margin” OR “Resection margins” OR “Safe margin” OR “OR Safe margins” OR “Specimen margin” OR “Specimen margins” OR “Surgical margin” OR “Surgical margins” OR “Tumor-free margin” OR “Tumor-free margins” OR “Tumour-free margin” OR “Tumour-free margins”):ti,ab,kw

Part 2: Diagnostic imaging
[mh “Diagnostic imaging”] OR [mh /DG]
OR
(“CT” OR “echography” OR “fluescence” OR “fluoscopy” OR “image” OR “images” OR “imaging” OR “microscopy” OR “MR” OR “MRI” OR “NIR” OR “OCT” OR “photography” OR “radiography” OR “radioisotope scanning” OR “roentgenography” OR “scintigraphy” OR “scintiphotography” OR “spectroscopy” OR “stroboscopy” OR “thermography” OR “tomography” OR “ultrasonography” OR “ultrasound” OR “X-ray” OR “X-rays”):ti,ab,kw

Part 3: Histology
[mh “Histology”] OR [mh “Pathology”] OR [mh /PA]
OR
(Histologic OR Histological OR Histologically OR Histology OR Histopathologic OR Histopathological OR Histopathologically OR Histopathology OR Pathologic OR Pathological OR pathologically OR Pathology):ti,ab,kw

2. Handling of duplicates

Duplicate articles were manually filtered using Endnote versions X8, Clarivate Analytics, Philadelphia, PA.

3. In- and exclusion criteria

Studies fulfilling all of the following inclusion criteria were considered eligible for this study:

- The aim to assess margins in patients undergoing oncologic surgery
- Use of an imaging technique, i.e. a technique that visually displays any structural of functional patterns of organs or tissues.
- Use of histology as reference standard
- The application of the imaging technique should be intended for use in the intraoperative setting
- The area of interest had to be fresh, i.e. not frozen or processed for H&E analysis, during the imaging procedure.
- Present original study data
- Time of initial publication is from the first of January 2013 until the end of December 2018

There were no exclusion criteria.

4. Quality assessment

The risk of bias and applicability of the studies was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) scoring system [1]. This is a validated tool for assessment of the methodological quality and applicability of diagnostic accuracy studies.
Four domains are distinguished:

- Patient selection, which describes the process of patient selection and included patients.
- Index test, describing the test under study and how it was conducted and interpreted.
- Reference standard, describing the used reference standard and how it was conducted and interpreted
- Flow and timing, describing the flow of patient in- and exclusion and the time interval between index test and reference standard.

One QUADAS form per study was filled out. If a study investigated multiple imaging techniques these were jointly scored on one form, since the investigated techniques

were usually similar. Besides the signaling questions provided in QUADAS, the following specific signaling questions were added for this review:

- Was the comparison of tumor with healthy samples avoided? If this was not the case, the goal of the imaging technique was considered to detect tumor instead of to assess margins. This was a cause for concerns regarding applicability. Furthermore, the use of tumor and healthy samples disturbs the normal distribution of positive and negative outcomes, which could result in non-representative number for diagnostic accuracy. Therefore the selection of patients, in this case samples, was considered as a source of bias.
- Was inclusion of benign tumors avoided? Some studies did not exclusively include malignancies, which was therefore a cause for concern that the included patients and setting did not match the review question. Studies with such designs were only included if histological proven cancer (e.g. brain tumors) before surgery was not a requirement by standard of care.

In case of inappropriate exclusions, e.g. exclusion of specimens in which the tumor could not be detected by the imaging technique, high risk was automatically scored both in the patient selection domain as well as in the flow and timing domain. This might have overestimated the total risk of bias in these studies.

5. Data synthesis

Data was extracted on patient and study characteristics, technical properties of the imaging technique, criteria describing the feasibility of the imaging technique and diagnostic accuracy.

Patient and study characteristics included:

- Title and authors
- Year of publication
- Study country, that is the country in which the study was conducted. If this was not mentioned, the author affiliations were used to determine study country.
- Cancer type
- T-classification
- Number of patients
- Type of surgery: open or laparoscopic. If the approach was not mentioned but the type of cancer evidently dictated a certain approach, for instance in breast cancer where an open approach is always employed, then this approach was recorded.
- Area of interest: surgical specimen, surgical cavity, surgical field, or combinations.
- Type of imaging technique
- IDEAL stage. IDEAL (Idea, Development, Exploration, Assessment, Long-term study) is a framework for describing the stages of innovation in surgery and other interventional procedures [2]. Their recommendations were followed to classify studies into one of the five stages of innovation in surgery. An additional criterion was added to better discriminate stage 2a studies from stage 2b. In case surgeons

made intraoperative decisions based on the imaging technique, the study was automatically classified as a stage 2b.

Technical properties of the imaging technique included:

- Scanning area or volume. This was the area or volume in mm^2 or mm^3 that was scanned in the study.
- Resolution in millimeters. If the resolution of the investigated technique was not described, it was derived from the combination of field of view and image matrix where possible. The resolution for fluorescence techniques, if not provided by the study authors, was reported as macroscopic, since the human eye or a surgical microscope was used to view imaging results. The (maximum) resolution of ultrasound, if not reported by the authors, was derived from the employed frequency using Abbé's diffraction limit.
- Penetration depth in millimeters. Being an inherent property of projectional radiography, and tomographic techniques, i.e. CT, MRI, PET, SPECT, their penetration depths were assumed to be sufficient for penetration of either the entire specimen or surgical cavity. The penetration depth of ultrasound was conservatively estimated at 20 millimeters.
- Acquisition time in minutes. This comprised the scanning, post-processing and reading time. However some studies reported only one or two of these or only reported "acquisition time" without further explanation, which might have limited the comparability of the acquisition times between imaging techniques.

Criteria describing the feasibility of the imaging technique in clinical practice adapted from Ha et al. [3]

- Continuous feedback. Technique allows for continuous margin assessment during surgery.
- Resolution comparable with histopathology. Technique has a spatial resolution that enables visualization of individual cells, i.e. 15 µm [3].
- Automatic acquisition. Automatic acquisition of the data in a standardized manner to eliminate sampling errors, ideally not requiring the need for a human operator
- Scanning of entre specimen or cavity. Technique is able to assess the complete resection specimen or surgical cavity
- Subsurface information. Technique allows clinicians to assess tissue structures situated at least 1mm below the surface of the sample.
- Non-destructive technique. Use of the technique is not destructive to the tissue in any way and therefore does not interfere with standard pathological work-up.
- Within intraoperative timescale. The technique does not substantially extend the time required for the surgical procedure. If scanning was performed within 30 minutes, as comparable to frozen section analysis, this was considered as feasible within the intraoperative timescale [4–6].
- Integration into surgical workflow. Use of the technique fits into the standard surgical workflow and does not substantially hinder it by needing extra room, personnel or different surgical techniques.

- Translatable to intraoperative action. The technique allows for the immediate intraoperative use of its outcomes, for instance an immediate additional resection in case of a positive or close margin.

Diagnostic accuracy:

- We distinguished between feasibility studies that calculated diagnostic accuracy based on the assessment of samples of tumor versus normal tissue and proof-of-concept studies that calculated diagnostic accuracy based on the assessment of inadequate margins. We further differentiated inadequate margins as either positive margins or positive and close margins.
- Handling of missed tumors. We reported whether proof-of-concept studies excluded specimens from the diagnostic accuracy analysis if their imaging technique was not able to detect the tumor in the specimen.
- Number of patients, true positive, true negative, false positive, false negative, positive reference standard and negative reference standard findings. These values were extracted from the study data or calculated from reported sensitivity and specificity values.
- Sensitivity, specificity, positive predictive value and negative predictive value. These values were extracted from the study data or calculated from reported true positive, true negative, false positive and false negative findings. In case there were discrepancies between reported and calculated diagnostic accuracy, we reported the data calculated from the original true positive, true negative, false positive and false negative findings.
- In case multiple studies used the same imaging technique in the same type of tumor, the possibility of performing meta-analysis was assessed.
- For proof-of-concept studies, the correct detection of a positive or close margin was considered a true positive finding, whereas correct detection of a negative margin was considered a true negative finding.

References

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Supplementary 2: Results

1. Systematic search

Full-text articles were excluded because they had different a domain, determinant or outcome (n=128), were not original studies (n=17), were double publications that used the same patients (n=11), were conference abstracts (n=15) or did not included human subjects (n =3). The domain, determinant and outcome category was further divided into:

- Domain (patients receiving oncologic surgery):
  - Wrong indication (n=24)
  - No fresh specimens (n=13)
- Determinant (intraoperative imaging versus histology)
  - No intraoperative feedback (n=49)
  - Wrong intervention (n=10)
  - Wrong comparator (n=23)
- Outcome (assessment of margins)
  - Wrong outcome (n=9)

NB: Giacomelli et al.[1] 2018, Yoshitake et al.[2] 2018, Yoshitake et al.[3] 2016, Tao et al.[4] 2014 and Cahill et al.[5] 2018, share one or multiple authorships and J.G. Fujimoto is last author of all 5 publications. It could not be assessed if these publications used the same patient data or not due to insufficient information provided in the paper. We contacted the corresponding author of these publications twice but unfortunately did not receive a response, therefore we included all five articles.

2. Quality assessment

Table 1 presents the results of the quality assessment. Risk of bias in patient selection was mainly caused by the non-consecutive or non-random inclusion of patients and the use of case-control designs. Risk of bias in the index test was mainly caused by a lack of blinding or unclear thresholds for test outcome. Lack of blinding was also the main cause of risk of bias in the reference standard category. In the flow and timing domain, selective inclusion of patients in the analysis was the main cause of risk of bias.

Applicability concerns in the index test were mainly caused by study designs in which tumor and healthy samples were used. In this case, the index test discriminated tumor from healthy tissue instead of positive or close margins from negative margins.

Table 1 Risk of bias and applicability concerns

Risk of bias			
	Low risk, n (%)	Unclear risk, n (%)	High risk, n (%)
Patient selection	24 (18)	69 (51)	41 (31)
Index test	78 (58)	25 (19)	31 (23)
Reference standard	111 (83)	21 (16)	2 (1)
Flow & timing	83 (62)	25 (19)	26 (19)
Applicability concerns			
	Low concern, n (%)	Unclear concern, n (%)	High concern, n (%)
Patient selection	129 (96)	3 (2)	2 (1)
Index test	98 (73)	33 (25)	3 (2)
Reference standard	131 (98)	3 (2)	0 (0)

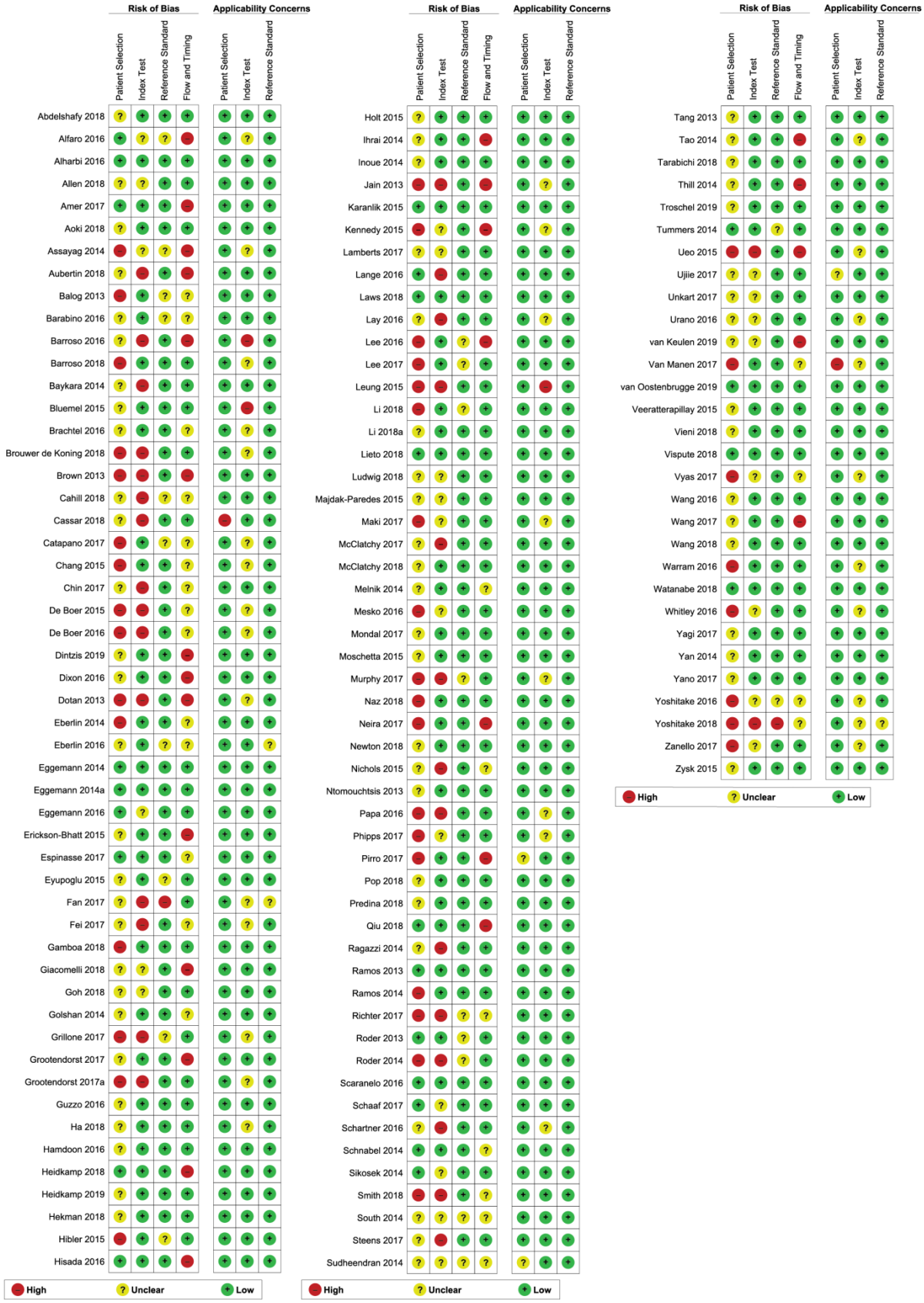


Figure 1 Risk of bias and applicability concerns for individual studies

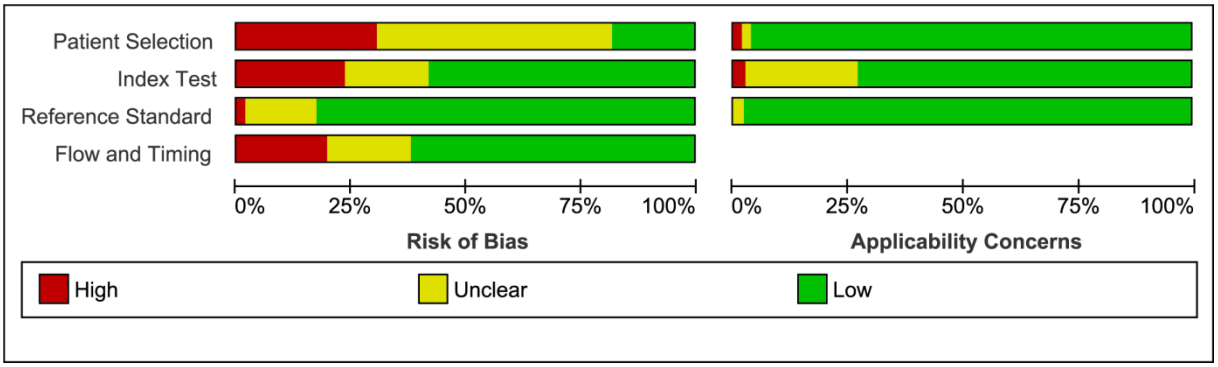


Figure 2 Overview of risk of bias and applicability concerns for all studies

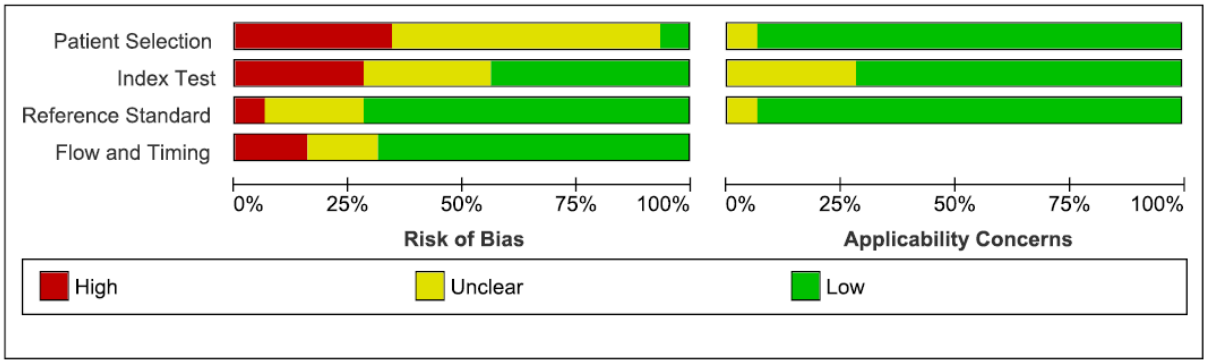


Figure 3 Quality assessment of IDEAL stage 1 studies

Study quality seemed to be dependent on the IDEAL stage, as stage 3 studies had higher overall quality than studies in other research stages. This can be seen in Figures.2-6.

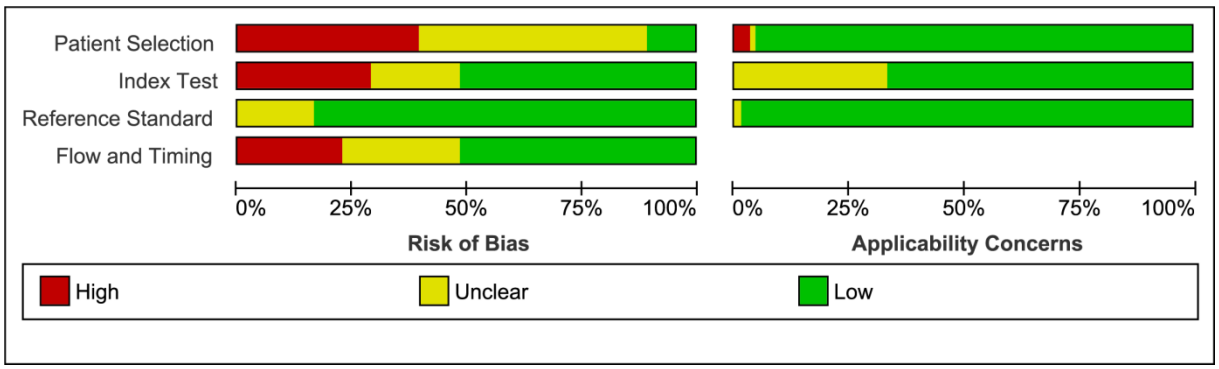


Figure 4 Quality assessment of IDEAL stage 2a studies

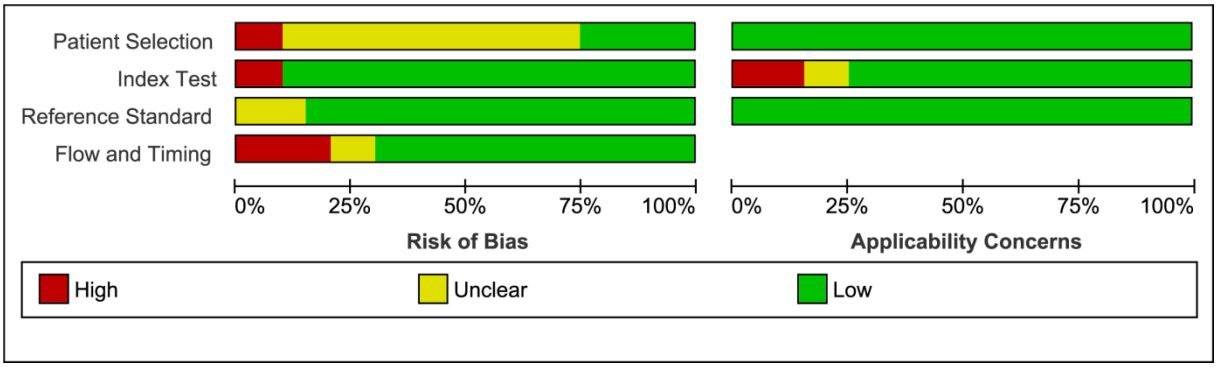


Figure 5 Quality assessment of IDEAL stage 2b studies

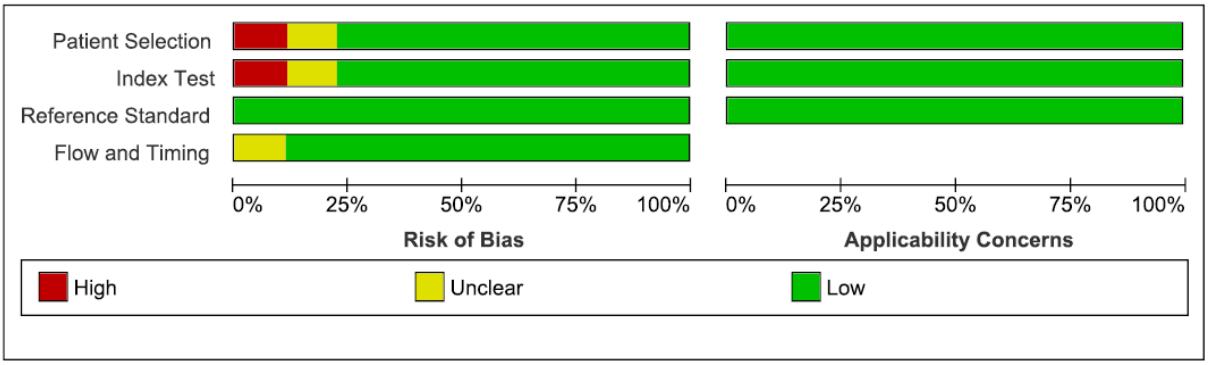


Figure 6 Quality assessment of IDEAL stage 3 studies

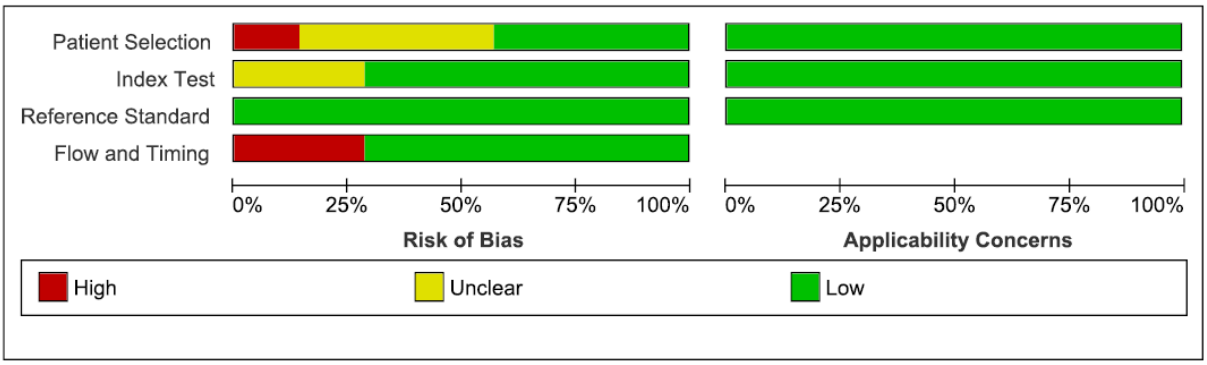


Figure 7 Quality assessment of IDEAL stage 4 studies

3. Patient and study characteristics

All studies were written in the English language, except for one that was written in Hebrew. The number of publications increased each year, as can be seen in Table 2.

Table 2 Number of studies per year of publication.

Year	N	%
2013	8	6
2014	20	15
2015	16	12
2016	24	18
2017	29	22
2018	32	24
2019	5	4

Table 3 Number of studies per country of origin

Country	N	%
USA	53	40
Netherlands	14	10
Germany	11	8
Japan	9	7
UK	8	6
France	7	5
Italy	5	4
Australia	4	3
Canada	4	3
Israel	3	2
China	2	1
Spain	2	1
Turkey	2	1
Egypt	1	1
Greece	1	1
Hong Kong	1	1
Hungary	1	1
India	1	1
Pakistan	1	1
Romania	1	1
Singapore	1	1
Slovenia	1	1
Sweden	1	1

Table 4 Number of studies per cancer type

Cancer type	N	%
Breast cancer	67	50
Head and neck cancer	12	9
Brain cancer	11	8
Kidney cancer	8	6
Multiple cancers	8	6
Prostate cancer	7	5
Sarcoma	4	3
Brain metastases	3	2
Lung cancer	3	2
Pancreatic cancer	3	2
Skin cancer	3	2
Colorectal cancer	1	1
Gastric cancer	1	1
Liver cancer	1	1
Liver metastases	1	1
Vulva cancer	1	1

Table 5 Number of investigated techniques per area of interest

Area of interest	N
Specimen	108
Cavity	14
Surgical field, cavity, specimen	9
Surgical field, cavity	6
Cavity, specimen	4
Surgical field, specimen	1
Surgical field	1

Table 6 Number of studies per IDEAL stage

IDEAL stage[6]	N	%
1 Idea: feasibility studies, few patients	32	24
2a Development: case series	66	49
2b Exploration: longer case series, intraoperative decision making	20	15
3 Assessment: RCT	9	7
4 Long-term study: registry, normal practice	7	5

Table 7 Number of investigated techniques stratified per imaging category

We have collected techniques under the term advanced microscopy, as those light-based instruments use relatively new or unconventional source configurations (such as geometrically- or spectrally encoded illuminations), relatively new or complex probe variants (multi-modal, endomicroscopy, or other), non-linear contrast mechanisms (two-photon and multi-photon) and confocal methods. Our definition is merely for the sake of grouping the techniques in this review. We have chosen to retain optical coherence tomography (OCT) as a separate category due to the relatively high number of studies using this method compared with the individual methods under advance microscopy.

Imaging category	**N**	**%**
Advanced microscopy	**18**	**13**
Non linear microscopy (NLM)	4	3
Fluorescence confocal microscopy (FCM)	3	2
Confocal laser endomicroscopy (CLE)	2	1
Multiphoton microscopy (MPM)	2	1
Structured illumination microscopy (SIM)	2	1
Confocal microscopy, two-photon microscopy, fluorescence life-time imaging	1	1
Reflectance confocal microscopy (RCM)	1	1
Microscopy with ultraviolet surface excitation (MUSE)	1	1
Probe-based confocal laser endomicroscopy (pCLE)	1	1
Spectrally encoded confocal microscopy (SECM)	1	1
Bio-impedance	**6**	**4**
Radio Frequency Spectroscopy	4	3
Bio-Impedance Spectroscopy	2	1
CT	**6**	**4**
Micro CT	4	3
Flat-panel volume computed tomography (fpVCT)	1	1
Micro-CT and spatial frequency domain imaging (SFDI)	1	1
Elastic scattering spectroscopy	**8**	**6**
Diffuse reflectance spectroscopy (DRS)	3	2
Elastic light single scattering spectroscopy (ELSSS)	2	1
Quantitative diffuse reflectance imaging (QDRI)	2	1
Light reflectance spectroscopy (LRS)	1	1
Fluorescence	**32**	**24**
Fluorescence	30	22
Fluorescence lifetime imaging (FLIm)	1	1
Fluorescence spectroscopy based hand-held probe	1	1
Hyperspectral imaging	**1**	**1**
Hyperspectral imaging (HSI)	1	1
Mass spectrometry	**5**	**3**
Desorption electrospray ionization mass spectroscopy imaging (DESI-MSI)	3	2
Rapid evaporative ionization mass spectrometry (REIMS)	1	1
Touch spray mass spectroscopy (TS-MS)	1	1

Imaging category	N	%
MRI	**10**	**7**
MRI	10	7
Nuclear medicine imaging	**3**	**2**
Cerenkov luminescence imaging (CLI)	1	1
Freehand single photon emission computed tomography (fhSPECT)	1	1
Specimen positron emission mammography (sPEM)	1	1
Optical coherence tomography	**13**	**10**
Optical coherence tomography (OCT)	7	5
Full field optical coherence tomography (FF-OCT)	3	2
Optical coherence elastography (OCE)	2	1
Optical coherence microelastography (OCME)	1	1
pH measurement	**1**	**1**
fiber-tip pH probe	1	1
Photoacoustic imaging	**2**	**1**
Multispectral optoacoustic tomography (MSOT)	1	1
Photoacoustic tomography (PAT)	1	1
Radiography	**13**	**10**
Specimen radiography	10	7
Digital breast tomosynthesis (DBT)	2	1
Full field digital mammography (FFDM)	1	1
Raman spectroscopy	**5**	**4**
Raman spectroscopy	3	2
Line-scan Raman microscopy	1	1
Raman encoded molecular imaging (REMI)	1	1
Terahertz imaging	**3**	**2**
Terahertz time-domain spectroscopy (THz-TDS)	1	1
Terahertz pulsed Imaging (TPI)	1	1
Terahertz pulsed imaging (TPI) + Wide-field polarization enhanced optical imaging	1	1
Ultrasound	**17**	**13**
Ultrasound	17	13

4. Diagnostic accuracy

Multiple studies investigated the same imaging technique in the same type of tumor, for example radiography and ultrasound in breast cancer. However, due to heterogeneity in clinical practice and tumor subtypes between studies it was not possible to conduct meta-analysis of the results.

References

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Supplementary Table 1 Patient and study characteristics

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
Advanced microscopy								
Brachtel et al. [1], 2016	USA	2a	35	Breast	-	Open	Spectrally encoded confocal microscopy	Specimen
Cahill et al. [2], 2018	USA	2a	-	Breast	-	-	Non linear microscopy (acridine orange and sulforhodamine 101)	Specimen
Chang et al. [3], 2015	UK	2a	50	Breast	-	Open	Probe-based confocal laser endomicroscopy (acriflavine hydrochloride)	Specimen
Espinasse et al. [4], 2017	France	2a	41	Skin	-	Open	Fluorescence confocal microscopy	Specimen
Maki et al. [5], 2017	Japan	2a	6	Liver	-	Laparoscopic	Confocal laser endomicroscopy	Specimen
Tao et al. [6], 2014	USA	2a	50	Breast	-	Open	Non linear microscopy (acridine orange)	Specimen
Vyas et al. [7], 2017	UK	2a	5	Breast	-	Open	Confocal laser endomicroscopy (methylene-blue)	Specimen
Wang et al. [8], 2016	USA	2a	24	Prostate	T2-T3	Laparoscopic	Structured illumination microscopy (acridine orange)	Specimen
Wang et al. [9], 2018	USA	2a	17	Kidney	T1-T2	Laparoscopic	Structured illumination microscopy (acridine orange)	Specimen
Yan et al. [10], 2014	China	2a	30	Rectum	stage I-III	-	Multiphoton microscopy	Specimen
Yoshitake et al. [11], 2016	USA	2a	-	Breast	-	-	Fluorescence confocal microscopy	Specimen
Yoshitake et al. [11], 2016	USA	2a	-	Breast	-	-	Multiphoton microscopy	Specimen
Giacomelli et al. [12], 2018	USA	1	-	Breast	-	Open	Non linear microscopy (acridine orange and sulforhodamine 101)	Specimen
Hibler et al. [13], 2015	USA	1	1	Skin	-	Open	Reflectance confocal microscopy	Cavity
Ragazzi et al. [14], 2014	Italy	1	30	Breast Colon Thyroid	-	-	Fluorescence confocal microscopy (acridine orange)	Specimen
Yoshitake et al. [15], 2018	USA	1	-	Breast Skin	-	-	Microscopy with ultraviolet surface excitation (propidium iodide and eosin yellow)	Specimen
Yoshitake et al. [15], 2018	USA	1	-	Breast Skin	-	-	Non linear microscopy (propidium iodide and eosin yellow)	Specimen
Zanello et al. [16], 2017	France	1	38	Brain metastases	-	-	Confocal microscopy Two-photon microscopy Fluorescence life-time imaging	Specimen
Bio-impedance								
Thill et al. [17], 2014	Germany	4	42	Breast	-	Open	Radio frequency spectroscopy	Specimen
Schnabel et al. [18], 2014	USA	3	596	Breast	-	Open	Radio frequency spectroscopy	Specimen

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
Dixon et al. [19], 2016	UK	2b	63	Breast	-	Open	Bio-impedance spectroscopy	Specimen
Melnik et al. [20], 2014	Israel	2b	64	Breast	-	Open	Radio frequency spectroscopy	Specimen
Dotan et al. [21], 2013	Israel	2a	49	Prostate	-	-	Radio frequency spectroscopy	Specimen
Murphy et al. [22], 2017	USA	2a	19	Prostate	-	Laparoscopic	Bio-impedance spectroscopy	Specimen
CT								
McClatchy et al. [23], 2018	USA	2a	32	Breast	-	Open	Micro CT	Specimen
Qiu et al. [24], 2018	Netherlands	2a	30	Breast	-	Open	Micro CT	Specimen
Tang et al. [25], 2013	USA	2a	6	Breast	-	Open	Micro CT	Cavity
Troschel et al. [26], 2019	USA	2a	21	Lung	-	-	Micro CT	Specimen
McClatchy et al. [27], 2017	USA	1	1	Breast	-	Open	Micro-CT and structured light imaging	Specimen
Schaaf et al. [28], 2017	Germany	1	47	Head and neck	-	Open	Flat-panel volume computed tomography	Specimen
Elastic scattering spectroscopy								
Baykara et al. [29], 2014	Turkey	2a	18	Prostate	-	-	Elastic light single scattering spectroscopy	Cavity
Brouwer de Koning et al. [30], 2018	Netherlands	2a	28	Head and neck	-	Open	Diffuse reflectance spectroscopy	Specimen
Brown et al. [31], 2013	USA	2a	99	Breast	-	Open	Quantitative diffuse reflectance imaging	Specimen
De Boer et al. [32], 2015	Netherlands	2a	16	Breast	-	Open	Diffuse reflectance spectroscopy	Specimen
De Boer et al. [33], 2016	Netherlands	2a	45	Breast	-	Open	Diffuse reflectance spectroscopy	Specimen
Grillone et al. [34], 2017	USA	2a	34	Head and neck	T1-T4	Open	Elastic light single scattering spectroscopy	Specimen
Lay et al. [35], 2016	USA	2a	50	Prostate	T1-T3	Laparoscopic	Light reflectance spectroscopy	Specimen
Nichols et al. [36], 2015	USA	2a	20	Breast	-	Open	Quantitative diffuse reflectance imaging	Specimen
Fluorescence								
Aoki et al. [37], 2018	Japan	2b	97	Liver Liver metastases	-	Laparoscopic	Fluorescence (indocyanine green)	Cavity
Catapano et al. [38], 2017	Italy	2b	48	Brain	-	Open	Fluorescence (fluorescein sodium)	Surgical field, cavity

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
Inoue et al. [39], 2014	Japan	2b	70	Liver Liver metastases	-	Both	Fluorescence (5-aminolevulinic acid)	Surgical field, cavity
Tummers et al. [40], 2014	Netherlands	2b	24	Breast	Stage 0-3	Open	Fluorescence (methylene blue)	Surgical field, cavity, specimen
Yano et al. [41], 2017	Japan	2b	8	Brain	-	Open	Fluorescence (5-aminolevulinic acid)	Cavity
Yano et al. [41], 2017	Japan	2b	8	Brain	-	Open	Fluorescence (fluorescein sodium)	Cavity
Dintzis et al. [42], 2019	USA	2a	23	Breast	-	Open	Fluorescence (Tozuleristide (BLZ-100))	Specimen
Lamberts et al. [43], 2017	Netherlands	2a	20	Breast	T1-T2	Open	Fluorescence (bevacizumab–IRDye800CW)	Surgical field, cavity, specimen
Lee et al. [44], 2017	USA	2a	13	Brain metastases	-	Open	Fluorescence (indocyanine green)	Surgical field, cavity
Lieto et al. [45], 2018	Italy	2a	9	Liver Liver metastases	-	Open	Fluorescence (indocyanine green)	Surgical field, cavity, specimen
Mondal et al. [46], 2017	USA	2a	3	Breast	-	Open	Fluorescence (indocyanine green)	Cavity, specimen
Neira et al. [47], 2017	USA	2a	32	Brain	WHO-IV	Open	Fluorescence (fluorescein sodium)	Surgical field
Newton et al. [48], 2018	USA	2a	20	Pancreas	-	Both	Fluorescence (indocyanine green)	Surgical field, cavity, specimen
Phipps et al. [49], 2017	USA	2a	14	Breast	-	Open	Fluorescence lifetime imaging	Specimen
Richter et al. [50], 2017	Sweden	2a	16	Brain	WHO III-IV	Open	Fluorescence spectroscopy based hand-held probe (5-aminolevulinic acid)	Cavity
Richter et al. [50], 2017	Sweden	2a	16	Brain	WHO III-IV	Open	Fluorescence (5-aminolevulinic acid)	Cavity
Smith et al. [51], 2018	USA	2a	15	Breast	-	Open	Fluorescence (LUM015)	Cavity, specimen
Yagi et al. [52], 2017	Japan	2a	16	Brain metastases	-	-	Fluorescence (5-aminolevulinic acid)	Surgical field, cavity
Barabino et al. [53], 2016	France	1	3	Liver metastases	-	Open	Fluorescence (indocyanine green)	Specimen
Eyupoglu et al. [54], 2015	Germany	1	3	Brain	-	Open	Fluorescence (5-aminolevulinic acid, indocyanine green)	Cavity
Guzzo et al. [55], 2016	USA	1	4	Kidney	T1	Open	Fluorescence (folate-fluorescein isothiocyanate)	Surgical field, cavity

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
Hekman et al. [56], 2018	Netherlands	1	15	Kidney	T1	Both	Fluorescence (111In-DOTA-girentuximab-IRDye800CW)	Surgical field, specimen
Holt et al. [57], 2015	USA	1	2	Soft tissue sarcoma	-	-	Fluorescence (indocyanine green)	Surgical field, cavity, specimen
Lee et al. [58], 2016	USA	1	15	Brain	WHO I-IV	Open	Fluorescence (indocyanine green)	Surgical field, cavity, specimen
Li et al. [59], 2018	China	1	14	Brain	WHO II-IV	Open	Fluorescence (68-Ga-IRDye800CW-bombesin)	Surgical field, cavity
Predina et al. [60], 2018	USA	1	1	Soft tissue sarcoma	-	Open	Fluorescence (indocyanine green)	Surgical field, cavity, specimen
Ueo et al. [61], 2015	Japan	1	5	Breast	-	Open	Fluorescence (gammaglutamyl hydroxymethyl rodamine green)	Specimen
Ujiie et al. [62], 2017	Canada	1	20	Lung	T1	Laparoscopic	Fluorescence (indocyanine green)	Specimen
Unkart et al. [63], 2017	USA	1	27	Breast	T1-T3	Open	Fluorescence (AVB-620)	Specimen
Van Keulen et al. [64], 2019	USA	1	8	Head and neck	T2-T3	Open	Fluorescence (panitumumab-IRDye800CW)	Specimen
Warram et al. [65], 2016	USA	1	3	Head and neck	-	Open	Fluorescence (cetuximab-IRDye800CW)	Specimen
Whitley et al. [66], 2016	USA	1	15	Breast Soft tissue sarcoma	-	Open	Fluorescence (LUM015)	Specimen
Hyperspectral imaging								
Fei et al. [67], 2017	USA	1	16	Head and neck	-	-	Hyperspectral imaging	Specimen
Mass spectrometry								
Alfaro et al. [68], 2016	USA	2a	21	Kidney	T1-T4	-	Touch spray mass spectroscopy	Specimen
Eberlin et al. [69], 2014	USA	2a	9	Stomach	-	-	Desorption electrospray ionization mass spectroscopy imaging	Specimen
Eberlin et al. [70], 2016	USA	2a	32	Pancreas	T1-T3	-	Desorption electrospray ionization mass spectroscopy imaging	Specimen
Pirro et al. [71], 2017	USA	2a	10	Brain	WHO I-IV	Open	Desorption electrospray ionization mass spectroscopy imaging	Cavity
Balog et al. [72], 2013	Hungary	1	81	Brain Breast Colorectal	-	-	Rapid evaporative ionization mass spectrometry	Specimen

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
				Liver Lung Stomach				
MRI								
Roder et al. [73], 2013	Germany	2b	22	Brain	WHO III-IV	Open	MRI	Cavity
Golshan et al. [74], 2014	USA	2a	8	Breast	-	Open	MRI	Cavity
Heidkamp et al. [75], 2018	Netherlands	2a	12	Prostate	T2-T3	Both	MRI	Specimen
Heidkamp et al. [76], 2019	Netherlands	2a	9	Vulva	FIGO 1A-3A	Open	MRI	Specimen
Mesko et al. [77], 2016	USA	2a	1	Soft tissue sarcoma	-	Open	MRI	Surgical field, cavity, specimen
Roder et al. [78], 2014	Germany	2a	45	Brain	WHO II-IV	Open	MRI	Cavity
Van Oostenbrugge et al. [79], 2019	Netherlands	2a	9	Kidney	-	Both	MRI	Specimen
Gamboa et al. [80], 2018	USA	1	1	Brain	-	Open	MRI	Cavity
Papa et al. [81], 2016	Israel	1	22	Breast	-	Open	MRI	Specimen
Steens et al. [82], 2017	Netherlands	1	10	Head and neck	T1-T4	Open	MRI	Specimen
Nuclear medicine imaging								
Bluemel et al. [83], 2015	Germany	2b	31	Breast	T1-T2	Open	Freehand single photon emission computed tomography (^{99m}Tc-nanocolloid)	Cavity, specimen
Grootendorst et al. [84], 2017	UK	1	12	Breast	-	Open	Cerenkov luminescence imaging (^{18}F-FDG)	Specimen
Watanabe et al. [85], 2018	Japan	1	16	Breast	stage 0-2a	Open	Specimen positron emission mammography (^{18}F-FDG)	Specimen
Optical coherence tomography								
Ha et al. [86], 2018	USA	2b	35	Breast	-	Open	Optical coherence tomography	Specimen
Allen et al. [87], 2018	Australia	2a	17	Breast	-	Open	Optical coherence microelastography	Specimen
Assayag et al. [88], 2014	France	2a	22	Breast	-	Open	Full field optical coherence tomography	Specimen
Chin et al. [89], 2017	Australia	2a	9	Breast	-	Open	Optical coherence elastography	Specimen
Erickson-Bhatt et al. [90], 2015	USA	2a	35	Breast	-	Open	Optical coherence tomography	Cavity, specimen

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
Hamdoon et al., [91], 2016	UK	2a	28	Head and neck	T1-T2	Open	Optical coherence tomography	Specimen
Jain et al. [92], 2013	USA	2a	13	Lung	-	-	Full field optical coherence tomography	Specimen
Kennedy et al. [93], 2015	Australia	2a	31	Breast	-	Open	Optical coherence elastography	Specimen
Ludwig et al. [94], 2018	USA	2a	15	Kidney	T1a-T3a	Both	Optical coherence tomography	Specimen
South et al. [95], 2014	USA	2a	8	Breast	-	Open	Optical coherence tomography	Specimen
Van Manen et al. [96], 2017	Netherlands	2a	29	Pancreas	-	Open	Full field optical coherence tomography	Specimen
Zysk et al. [97], 2015	USA	2a	46	Breast	-	Open	Optical coherence tomography	Specimen
Sudheendran et al. [98], 2014	USA	1	-	Soft tissue sarcoma	-	-	Optical coherence tomography	Specimen
pH measurement								
Schartner et al. [99], 2016	Australia	1	8	Breast Skin	-	Open	Fiber-tip pH probe	Specimen
Photoacoustic imaging								
Li et al. [100], 2018a	USA	2a	66	Breast	-	Open	Photoacoustic tomography	Specimen
Goh et al. [101], 2018	Singapore	1	1	Breast	-	Open	Multispectral optoacoustic tomography	Specimen
Radiography								
Hisada et al. [102], 2016	Japan	4	174	Breast	T1-T4	Open	Specimen radiography	Specimen
Laws et al. [103], 2018	Canada	4	588	Breast	-	Open	Specimen radiography	Specimen
Majdak-Paredes et al. [104], 2015	UK	4	98	Breast	-	Open	Specimen radiography	Specimen
Naz et al. [105], 2018	Pakistan	4	52	Breast	-	Open	Specimen radiography	Specimen
Lange et al. [106], 2016	Germany	3	91	Breast	-	Open	Specimen radiography	Specimen
Amer et al. [107] , 2017	Germany	2b	102	Breast	-	Open	Digital breast tomosynthesis	Specimen
Amer et al. [107], 2017	Germany	2b	102	Breast	-	Open	Full field digital mammography	Specimen
Ihrai et al. [108], 2014	France	2b	170	Breast	-	Open	Specimen radiography	Specimen
Leung et al. [109], 2015	Hong Kong	2b	23	Breast	-	Open	Specimen radiography	Specimen
Ntomouchtsis et al. [110], 2013	Greece	2b	16	Head and neck	-	Open	Specimen radiography	Specimen

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
Pop et al. [111], 2018	Romania	2b	84	Breast	T1-T3	Open	Specimen radiography	Specimen
Urano et al. [112], 2016	Japan	2a	65	Breast	Tis-T2	Open	Digital breast tomosynthesis	Specimen
Urano et al. [112], 2016	Japan	2a	65	Breast	Tis-T2	Open	Specimen radiography	Specimen
Raman spectroscopy								
Barroso et al. [113], 2016	Netherlands	2b	20	Head and neck	T1-T4	Open	Raman spectroscopy	Specimen
Aubertin et al. [114], 2018	Canada	2a	18	Prostate	-	-	Raman spectroscopy	Specimen
Barroso et al. [115], 2018	Netherlands	2a	22	Head and neck	T0-T4a	Open	Raman spectroscopy	Specimen
Wang et al. [116], 2017	USA	2a	29	Breast	-	Open	Raman encoded molecular imaging	Specimen
Sudheendran et al. [98], 2014	USA	1	-	Soft tissue sarcoma	-	-	Line scan raman spectroscopy	Specimen
Terahertz imaging								
Cassar et al. [117], 2018	France	2a	11	Breast	-	Open	Terahertz time-domain spectroscopy	Specimen
Grootendorst et al. [118], 2017a	UK	2a	32	Breast	-	Open	Terahertz pulsed imaging	Specimen
Fan et al. [119], 2017	USA	1	8	Skin	-	Open	Terahertz pulsed imaging + wide-field polarization enhanced optical imaging	Specimen
Ultrasound								
Laws et al. [103], 2018	Canada	4	588	Breast	-	Open	Ultrasound	Specimen
Sikosek et al. [120], 2014	Slovenia	4	125	Breast	-	Open	Ultrasound	Specimen
Vieni, et al. [121], 2018	Italy	4	842	Breast	T1-T3	Open	Ultrasound	Specimen
Eggemann et al. [122], 2014	Germany	3	198	Breast	-	Open	Ultrasound	Specimen
Eggemann et al. [123], 2016	Germany	3	158	Breast	-	Open	Ultrasound	Specimen
Karanlik et al. [124], 2015	Turkey	3	84	Breast	-	Open	Ultrasound	Specimen
Moschetta et al. [125], 2015	Italy	3	132	Breast	-	Open	Ultrasound	Specimen
Ramos et al. [126], 2013	Spain	3	223	Breast	UICC I-III	Open	Ultrasound	Specimen
Ramos et al. [127], 2014	Spain	3	58	Breast	T1-T3	Open	Ultrasound	Specimen
Vispute et al. [128], 2018	India	3	68	Breast	T1-T2	Open	Ultrasound	Specimen
Abdelshafy et al. [129], 2018	Egypt	2b	60	Breast	T1-T2	Open	Ultrasound	Surgical field,

Study	Country	IDEAL-stage	Number of patients	Cancer type	T-classification	Type of surgery	Imaging technique (fluorescent/contrast agent)	Area of interest
								cavity, specimen
Alharbi et al. [130], 2016	France	2b	147	Kidney	T1-T3	Both	Ultrasound	Specimen
Eggemann et al. [131], 2014a	Germany	2b	147	Breast	-	Open	Ultrasound	Specimen
Pop et al. [111], 2018	Romania	2b	84	Breast	T1-T3	Open	Ultrasound	Specimen
Veeratterapillay et al. [132], 2015	UK	2b	45	Kidney	T1; T3	Both	Ultrasound	Specimen
Scaranelo et al. [133], 2016	Canada	2a	25	Breast	-	Open	Ultrasound	Specimen
Tarabichi et al. [134], 2018	USA	2a	12	Tongue	T1-T2	Open	Ultrasound	Specimen

Supplementary table 2 Technical properties.

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Advanced microscopy							
Brachtel et al. [1], 2016	Spectrally encoded confocal microscopy	Breast	2a	10×4.8 mm	0.0013	0.2	10-15
Cahill et al. [2], 2018	Non linear microscopy (acridine orange and sulforhodamine 101)	Breast	2a	0.49×0.49 mm	0.0005	0.1	Near real time
Chang et al. [3], 2015	Probe-based confocal laser endomicroscopy (acriflavine hydrochloride)	Breast	2a	10×10 mm	0.001	0.06	3 (1-6)
Espinasse et al. [4], 2017	Fluorescence confocal microscopy	Skin	2a	Entire specimen	0.001	0.2	15
Maki et al. [5], 2017	Confocal laser endomicroscopy	Liver	2a	0.6 mm	0.0035	0.05	Near real time
Tao et al. [6], 2014	Non linear microscopy (acridine orange)	Breast	2a	1 cross-sectional plane of bread loafed specimen	0.00044	0.2	8.9 acquisition + 1.14 reading
Vyas et al. [7], 2017	Confocal laser endomicroscopy (methylene-blue)	Breast	2a	10×10 mm	0.0022	-	-
Wang et al. [8], 2016	Structured illumination microscopy (acridine orange)	Prostate	2a	4 margin surfaces	0.0013	-	60
Wang et al. [9], 2018	Structured illumination microscopy (acridine orange)	Kidney	2a	1 margin surface	0.0013	-	30
Yan et al. [10], 2014	Multiphoton microscopy	Rectum	2a	Distal margin	-	Several hundred µm	5
Yoshitake et al. [11], 2016	Fluorescence confocal microscopy	Breast	2a	5×5 mm	0.00044	0.075	-
Yoshitake et al. [11], 2016	Multiphoton microscopy	Breast	2a	5×5 mm	0.00048	0.075	-
Giacomelli et al. [12], 2018	Non linear microscopy (acridine orange and sulforhodamine 101)	Breast	1	Specimen slices	0.001	-	-
Hibler et al. [13], 2015	Reflectance confocal microscopy	Skin	1	Entire cavity	-	-	-
Ragazzi et al. [14], 2014	Fluorescence confocal microscopy (acridine orange)	Breast Colon Thyroid	1	12×12×3 mm	0.002	0.25	9
Yoshitake et al. [15], 2018	Microscopy with ultraviolet surface excitation (propidium iodide and eosin yellow)	Breast Skin	1	50 mm^2	-	Few µm	-
Yoshitake et al. [15], 2018	Non linear microscopy (propidium iodide and eosin yellow)	Breast Skin	1	50 mm^2	-	Few µm	-
Zanello et al. [16], 2017	Confocal microscopy Two-photon microscopy Fluorescence life-time imaging	Brain metastases	1	-	0.5	-	5-10
Bio-impedance							

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Thill et al. [17], 2014	Radio frequency spectroscopy	Breast	4	Entire specimen	-	-	5
Schnabel et al. [18], 2014	Radio frequency spectroscopy	Breast	3	Entire specimen	⌀7	-	5
Dixon et al. [19], 2016	Bio-impedance spectroscopy	Breast	2b	Entire specimen	-	3	5
Melnik et al. [20], 2014	Radio frequency spectroscopy	Breast	2b	Entire specimen	-	-	Few minutes
Dotan et al. [21], 2013	Radio frequency spectroscopy	Prostate	2a	Apex + bladder neck	0.8	0.2	1.3-4
Murphy et al. [22], 2017	Bio-impedance spectroscopy	Prostate	2a	116-154 mm^2	-	-	-
CT							
McClatchy et al. [23], 2018	Micro CT	Breast	2a	120x120x30 mm	0.15	Entire specimen	2
Qiu et al. [24], 2018	Micro CT	Breast	2a	Entire specimen	0.004-0.05	Entire specimen	13.5
Tang et al. [25], 2013	Micro CT	Breast	2a	Entire cavity	-	Entire specimen	14
Troschel et al. [26], 2019	Micro CT	Breast	2a	Entire specimen	0.055	Entire specimen	13
McClatchy et al. [27], 2017	Micro-CT and structured light imaging	Breast	1	1 slice	1	Entire specimen	14
Schaaf et al. [28], 2017	Flat-panel volume computed tomography	Head and neck	1	Entire specimen	0.23	Entire specimen	16
Elastic scattering spectroscopy							
Baykara et al. [29], 2014	Elastic light single scattering spectroscopy	Prostate	2a	3-4 mm^2	0.1	-	Near real time
Brouwer de Koning et al. [30], 2018	Diffuse reflectance spectroscopy	Head and neck	2a	6.6 times 1 mm	1	0.4-0.7	-
Brown et al. [31], 2013	Quantitative diffuse reflectance imaging	Breast	2a	2,000-4,000 mm^2	5	2	20-25
De Boer et al. [32], 2015	Diffuse reflectance spectroscopy	Breast	2a	1-3 mm^3	1.75	-	20
De Boer et al. [33], 2016	Diffuse reflectance spectroscopy	Breast	2a	Several mm^3	-	-	-
Grillone et al. [34], 2017	Elastic light single scattering spectroscopy	Head and neck	2a	⌀ 2 mm	2	0.5	Near real time
Lay et al. [35], 2016	Light reflectance spectroscopy	Prostate	2a	⌀1 mm	1	2	-
Nichols et al. [36], 2015	Quantitative diffuse reflectance imaging	Breast	2a	3,400 mm^2	0.75	2	28
Fluorescence							
Aoki et al. [37], 2018	Fluorescence (indocyanine green)	Liver Liver metastases	2b	Entire cavity	-	5	Near real time

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Catapano et al. [38], 2017	Fluorescence (fluorescein sodium)	Brain	2b	Entire surgical field + cavity	1	-	Near real time
Inoue et al. [39], 2014	Fluorescence (5-aminolevulinic acid)	Liver Liver metastases	2b	Entire surgical field + cavity	-	-	Near real time
Tummers et al. [40], 2014	Fluorescence (methylene blue)	Breast	2b	Entire surgical field + cavity + specimen	-	5	-
Yano et al. [41], 2017	Fluorescence (5-aminolevulinic acid)	Brain	2b	Entire surgical field	-	-	Near real time
Yano et al. [41], 2017	Fluorescence (fluorescein sodium)	Brain	2b	Entire surgical field	-	-	Near real time
Dintzis et al. [42], 2019	Fluorescence (Tozuleristide (BLZ-100))	Breast	2a	Entire specimen	-	-	-
Lamberts et al. [43], 2017	Fluorescence (bevacizumab–IRDye800CW)	Breast	2a	Entire surgical field + cavity + specimen	0.03	-	-
Lee et al. [44], 2017	Fluorescence (indocyanine green)	Brain metastases	2a	Entire surgical field + cavity	-	-	Near real time
Lieto et al. [45], 2018	Fluorescence (indocyanine green)	Liver Liver metastases	2a	Entire surgical field + cavity + specimen	-	8	20 (10-30)
Mondal et al., [46], 2017	Fluorescence (indocyanine green)	Breast	2a	Entire cavity + specimen	-	-	-
Neira et al. [47], 2017	Fluorescence (fluorescein sodium)	Brain	2a	Entire surgical field	-	-	Near real time
Newton et al. [48], 2018	Fluorescence (indocyanine green)	Pancreas	2a	Entire surgical field + cavity + specimen	-	-	-
Phipps et al. [49], 2017	Fluorescence lifetime imaging	Breast	2a	20×20×4-5 mm	-	0.3	Within minutes
Richter et al. [50], 2017	Fluorescence spectroscopy based hand-held probe (5-aminolevulinic acid)	Brain	2a	Point measurements	⌀ 2	-	-
Richter et al. [50]2017	Fluorescence (5-aminolevulinic acid)	Brain	2a	125×143 mm	-	-	Near real time
Smith et al. [51], 2018	Fluorescence (LUM015)	Breast	2a	-	-	-	1s per ⌀ 26 mm
Yagi et al. [52], 2017	Fluorescence (5-aminolevulinic acid)	Brain metastases	2a	Entire surgical field + cavity	-	-	-
Barabino et al. [53], 2016	Fluorescence (indocyanine green)	Liver metastases	1	Specimen slices	-	-	Near real time
Eyupoglu et al. [54], 2015	Fluorescence (5-aminolevulinic acid, indocyanine green)	Brain	1	Entire cavity	-	-	Near real time
Guzzo et al. [55], 2016	Fluorescence (folate-fluorescein isothiocyanate)	Kidney	1	Entire surgical field + cavity	-	-	-
Hekman et al. [56], 2018	Fluorescence (111In-DOTA-girentuximab-IRDye800CW)	Kidney	1	Entire surgical field + specimen	-	-	-
Holt et al. [57], 2015	Fluorescence (indocyanine green)	Soft tissue sarcoma	1	Entire surgical field + cavity + specimen	-	-	-

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Lee et al. [58], 2016	Fluorescence (indocyanine green)	Brain	1	Entire surgical field + cavity + specimen	-	-	Near real time
Li et al. [59], 2018	Fluorescence (68-Ga-IRDye800CW-bombesin)	Brain	1	Entire surgical field + cavity	-	-	Near real time
Predina et al. [60], 2018	Fluorescence (indocyanine green)	Soft tissue sarcoma	1	Entire surgical field + specimen + cavity	-	-	11
Ueo et al. [61], 2015	Fluorescence (gammaglutamyl hydroxymethyl rodamine green)	Breast	1	Entire specimen	-	-	-
Ujiie et al. [62], 2017	Fluorescence (indocyanine green)	Lung	1	Entire specimen	-	-	-
Unkart et al. [63], 2017	Fluorescence (AVB-620)	Breast	1	Entire specimen	-	-	Near real time
Van Keulen et al. [64], 2019	Fluorescence (panitumumab-IRDye800CW)	Head and neck	1	Entire specimen	0.125	6	7
Warram et al. [65], 2016	Fluorescence (cetuximab-IRDye800CW)	Head and neck	1	45 punch biopsies (⌀ 4 mm)	-	10	-
Whitley et al. [66], 2016	Fluorescence (LUM015)	Breast Soft tissue sarcoma	1	Samples	-	-	-
Hyperspectral imaging							
Fei et al. [67], 2017	Hyperspectral imaging	Head and neck	1	Parts of the specimen	-	-	-
Mass spectrometry							
Alfaro et al. [68], 2016	Touch spray mass spectroscopy	Kidney	2a	3×3×3 mm	-	n/a	20
Eberlin et al. [69], 2014	Desorption electrospray ionization mass spectroscopy imaging	Pancreas	2a	20×20 mm	0.2	n/a	-
Eberlin et al. [70], 2016	Desorption electrospray ionization mass spectroscopy imaging	Stomach	2a	20×20 mm	0.2	n/a	120
Pirro et al. [71], 2017	Desorption electrospray ionization mass spectroscopy imaging	Brain	2a	Biopsies	-	n/a	3/sample
Balog et al. [72], 2013	Rapid evaporative ionization mass spectrometry	Brain Breast Colorectal Liver Lung Stomach	1	-	1	n/a	1 minute for 24 mm
MRI							
Roder et al. [73], 2013	MRI	Brain	2b	Entire cavity	1.8	Entire cavity	25
Golshan et al. [74], 2014	MRI	Breast	2a	Entire cavity	-	Entire breast	-

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Heidkamp et al. [75], 2018	MRI	Prostate	2a	Entire specimen	0.13	Entire specimen	90
Heidkamp et al. [76], 2019	MRI	Vulva	2a	Entire specimen	0.16	Entire specimen	75
Mesko et al. [77], 2016	MRI	Soft tissue sarcoma	2a	Entire surgical field + cavity + specimen	-	Entire specimen	45
Roder et al. [78], 2014	MRI	Brain	2a	15×15×15 mm	15×15×15	Entire cavity	40
Van Oostenbrugge et al. [79], 2019	MRI	Kidney	2a	Entire specimen	0.19	Entire specimen	80
Gamboa et al. [80], 2018	MRI	Brain	1	Biopsies	-	-	-
Papa et al. [81], 2016	MRI	Breast	1	2-5 punch biopsies (⌀6 × 2-5 mm)	-	Entire sample	0.16
Steens et al. [82], 2017	MRI	Head and neck	1	Entire specimen	0.2	Entire specimen	90
Nuclear medicine imaging							
Bluemel et al. [83], 2015	Freehand single photon emission computed tomography (^{99m}Tc-nanocolloid)	Breast	2b	Entire cavity + specimen	5	-	1.5 (0.9-2.9)
Grootendorst et al. [84], 2017	Cerenkov luminescence imaging (^{18}F-FDG)	Breast	1	1 cross-sectional plane	1.25	-	5 imaging + 2 reading
Watanabe et al. [85], 2018	Specimen positron emission mammography (^{18}F-FDG)	Breast	1	Entire specimen	2.5	Entire specimen	5-10
Optical coherence tomography							
Ha et al. [86], 2018	Optical coherence tomography	Breast	2b	10×10×2.5 - 14×14×3 mm	0.015-0.020	2	1 - 1.5
Allen et al. [87], 2018	Full field optical coherence tomography	Breast	2a	2 margins	0.013	n/a	210[1]
Assayag et al. [88], 2014	Full field optical coherence tomography	Breast	2a	225 mm^2	0.001	0.2	10
Chin et al. [89], 2017	Optical coherence elastography	Breast	2a	6×6 mm	0.011	3	-
Erickson-Bhatt et al. [90], 2015	Optical coherence tomography	Breast	2a	Entire cavity + specimen	0.009	-	Near real time
Hamdoon et al., [91], 2016	Optical coherence tomography	Head and neck	2a	Entire specimen	0.0053	1.5	Near real time

[1] 30 minutes acquisition + 180 minutes post-processing

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Jain et al. [92], 2013	Full field optical coherence tomography	Lung	2a	3×3×0.5 mm	0.0015	0.06	7
Kennedy et al. [93], 2015	Optical coherence elastography	Breast	2a	15×15×5 mm	0.011	0.3	16
Ludwig et al. [94], 2018	Optical coherence tomography	Kidney	2a	-	0.02	2-3	Near real time
South et al. [95], 2014	Optical coherence tomography	Breast	2a	30×20 mm	0.009	-	Near real time
Van Manen et al. [96], 2017	Full field optical coherence tomography	Pancreas	2a	⌀25 mm	0.0015	Several mm	30
Zysk et al. [97], 2015	Optical coherence tomography	Breast	2a	Entire specimen	0.015	2-3	Near real time
Sudheendran et al. [98], 2014	Optical coherence tomography	Soft tissue sarcoma	1	-	0.008	-	-
pH measurement							
Schartner et al. [99], 2016	Fiber-tip pH probe	Breast Skin	1	11-20 point measurements	0.2	-	11-20
Photoacoustic imaging							
Li et al. [100], 2018a	Photoacoustic tomography	Breast	2a	Entire specimen	0.2	6	10
Goh et al. [101], 2018	Multispectral optoacoustic tomography	Breast	1	Entire specimen	0.15	10-30	20
Radiography							
Hisada et al. [102], 2016	Specimen radiography	Breast	4	Entire specimen	0.1	Entire specimen	-
Laws et al. [103], 2018	Specimen radiography	Breast	4	Entire specimen	-	Entire specimen	-
Majdak-Paredes et al. [104], 2015	Specimen radiography	Breast	4	Entire specimen	-	Entire specimen	0.5
Naz et al. [105], 2018	Specimen radiography	Breast	4	Entire specimen	-	Entire specimen	-
Lange et al. [106], 2016	Specimen radiography	Breast	3	Entire specimen	-	Entire specimen	-
Amer et al. [107] , 2017	Digital breast tomosynthesis	Breast	2b	Entire specimen	1	Entire specimen	-
Amer et al. [107], 2017	Full field digital mamography	Breast	2b	Entire specimen	1	Entire specimen	-
Ihrai et al. [108], 2014	Specimen radiography	Breast	2b	Entire specimen	0.02	Entire specimen	3.5
Leung et al. [109], 2015	Specimen radiography	Breast	2b	Entire specimen	-	Entire specimen	-
Ntomouchtsis et al. [110], 2013	Specimen radiography	Head and neck	2b	Entire specimen	-	Entire specimen	20
Pop et al. [111], 2018	Specimen radiography	Breast	2b	Entire specimen	-	Entire specimen	-

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Urano et al. [112], 2016	Digital breast tomosynthesis	Breast	2a	Entire specimen	-	Entire specimen	-
Urano et al. [112], 2016	Specimen radiography	Breast	2a	Entire specimen	-	Entire specimen	-
Raman spectroscopy							
Barroso et al. [113], 2016	Raman spectroscopy	Head and neck	2b	240 mm^2	0.3	-	60
Aubertin et al. [114], 2018	Raman spectroscopy	Prostate	2a	10-50 point measurements of ⌀ 0.5 mm	2.1	-	Near real time (1 point measurement)
Barroso et al. [115], 2018	Raman spectroscopy	Head and neck	2a	128.7 mm^2	1	-	60
Wang et al. [116], 2017	Raman encoded molecular imaging	Breast	2a	-	0.5	-	15
Sudheendran et al. [98], 2014	Line scan raman spectroscopy	Soft tissue sarcoma	1	-	0.0006 × 0.0008	-	-
Terahertz imaging							
Cassar et al. [117], 2018	Terahertz time-domain spectroscopy	Breast	2a	-	0.7	-	45-60
Grootendorst et al. [118], 2017a	Terahertz pulsed imaging	Breast	2a	15×2 mm	-	-	1.5
Fan et al. [119], 2017	Terahertz pulsed imaging + wide-field polarization enhanced optical imaging	Skin	1	18×14 mm	0.012	-	30-90
Ultrasound							
Laws et al. [103], 2018	Ultrasound	Breast	4	Enitire specimen	-	20	-
Sikosek et al. [120], 2014	Ultrasound	Breast	4	Entire specimen	0.075-0.05	20	-
Vieni, et al. [121], 2018	Ultrasound	Breast	4	Entire specimen	0.0625-0.075	20	-
Eggemann et al. [122], 2014	Ultrasound	Breast	3	Entire specimen	0.05	20	Near real time
Eggemann et al. [123], 2016	Ultrasound	Breast	3	Entire specimen	0.05	20	-
Karanlik et al. [124], 2015	Ultrasound	Breast	3	Entire specimen	0.075	20	Near real time
Moschetta et al. [125], 2015	Ultrasound	Breast	3	Entire specimen	-	-	-
Ramos et al. [126], 2013	Ultrasound	Breast	3	Entire specimen	0.125-0.0625	20	-
Ramos et al. [127], 2014	Ultrasound	Breast	3	Entire specimen	-	-	-

Study	Imaging technique (fluorescent/contrast)	Cancer type	IDEAL –stage	Scanning volume/area	Resolution [mm]	Penetration depth [mm]	Acquisition time [min]
Vispute et al. [128], 2018	Ultrasound	Breast	3	Entire specimen	0.058-0.125	20	-
Abdelshafy et al. [129], 2018	Ultrasound	Breast	2b	Entire surgical field + specimen + cavity	0.1-0.0625	-	-
Alharbi et al. [130], 2016	Ultrasound	Kidney	2b	Entire specimen	0.06-0.15	20	1
Eggemann et al. [131], 2014a	Ultrasound	Breast	2b	Entire specimen	0.05	20	Near real time
Pop et al. [111], 2018	Ultrasound	Breast	2b	Entire specimen	0.0625	20	-
Veeratterapillay et al. [132], 2015	Ultrasound	Kidney	2b	Entire specimen	0.05	20	-
Scaranelo et al. [133], 2016	Ultrasound	Breast	2a	Entire specimen	0.10-0.04	20	-
Tarabichi et al. [134], 2018	Ultrasound	Head and neck	2a	Entire specimen	-	-	-

Supplementary table 3 Feasibility in clinical practice

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA [2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale [3]	Integration into surgical workflow	Translatable to intraoperative action
Advanced microscopy												
Brachtel et al. [1], 2016	Spectrally encoded confocal microscopy	Breast	2a	No	Yes	No	No	No	Yes	Yes	No	No
Cahill et al. [2], 2018	Non linear microscopy (acridine orange and sulforhodamine 101)	Breast	2a	No	Yes	No	No	No	Yes	-	No	No
Chang et al. [3], 2015	Probe-based confocal laser endomicroscopy (acriflavine hydrochloride)	Breast	2a	No	Yes	No	No	No	Yes	Yes	Yes	No
Espinasse et al. [4], 2017	Fluorescence confocal microscopy	Skin	2a	No	Yes	Yes	Yes	No	Yes	Yes	-	-
Maki et al. [5], 2017	Confocal laser endomicroscopy	Liver	2a	No	Yes	No	No	No	-	-	No	-
Tao et al. [6], 2014	Non linear microscopy (acriflavine hydrochloride)	Breast	2a	No	Yes	Yes	No	No	Yes	Yes	No	-
Vyas et al. [7], 2017	Confocal laser endomicroscopy (methylene-blue)	Breast	2a	No	Yes	Yes	No	No	Yes	-	No	No
Wang et al. [8], 2016	Structured illumination microscopy (acridine orange)	Prostate	2a	No	Yes	Yes	No	-	Yes	No	No	No
Wang et al. [9], 2018	Structured illumination microscopy (acridine orange)	Kidney	2a	No	Yes	Yes	Yes	-	Yes	Yes	No	-
Yan et al. [10], 2014	Multiphoton microscopy	Rectum	2a	No	-	-	No	No	Yes	Yes	No	No
Yoshitake et al. [11], 2016	Fluorescence confocal microscopy	Breast	2a	No	Yes	-	No	No	-	-	-	-
Yoshitake et al. [11], 2016	Multiphoton microscopy	Breast	2a	No	Yes	-	No	No	-	-	-	-
Giacomelli et al. [12], 2018	Non linear microscopy (acridine orange and sulforhodamine 101)	Breast	1	No	Yes	No	No	No	Yes	Yes	No	No
Hibler et al. [13], 2015	Reflectance confocal microscopy	Skin	1	No	-	Yes	Yes	-	Yes	Yes	Yes	No
Ragazzi et al. [14], 2014	Fluorescence confocal microscopy (acridine orange)	Breast Colon Thyroid	1	No	Yes	Yes	No	No	-	Yes	No	-

[2] i.e. 15µm or 0.015 mm

[3] i.e. 30 minutes

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA[2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale[3]	Integration into surgical workflow	Translatable to intraoperative action
Yoshitake et al. [15], 2018	Microscopy with ultraviolet surface excitation (propidium iodide and eosin yellow)	Breast Skin	1	No	-	-	No	No	-	-	-	No
Yoshitake et al. [15], 2018	Non linear microscopy (propidium iodide and eosin yellow)	Breast Skin	1	No	-	-	No	No	-	-	-	No
Zanello et al. [16], 2017	Confocal microscopy Two-photon microscopy Fluorescence life-time imaging	Brain metastases	1	No	No	Yes	No	-	Yes	Yes	-	No
Bio-impedance												
Thill et al. [17], 2014	Radio frequency spectroscopy	Breast	4	No	No	No	Yes	-	Yes	Yes	Yes	Yes
Schnabel et al. [18], 2014	Radio frequency spectroscopy	Breast	3	No	No	No	Yes	-	Yes	Yes	Yes	Yes
Dixon et al. [19], 2016	Bio-impedance spectroscopy	Breast	2b	No	-	No	Yes	Yes	Yes	Yes	Yes	Yes
Melnik et al. [20], 2014	Radio frequency spectroscopy	Breast	2b	No	-	No	Yes	-	Yes	Yes	Yes	Yes
Dotan et al. [21], 2013	Radio frequency spectroscopy	Prostate	2a	No	No	No	No	No	Yes	Yes	Yes	No
Murphy et al. [22], 2017	Bio-impedance spectroscopy	Prostate	2a	No	No	No	No	-	-	No	-	-
CT												
McClatchy et al. [23], 2018	Micro CT	Breast	2a	No	No	Yes	Yes	Yes	Yes	Yes	No	No
Qiu et al. [24], 2018	Micro CT	Breast	2a	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Tang et al. [25], 2013	Micro CT	Breast	2a	No	No	Yes	Yes	Yes	Yes	Yes	No	No
Troschel et al. [26], 2019	Micro CT	Breast	2a	No	No	Yes	Yes	Yes	Yes	Yes	No	No
McClatchy et al. [27], 2017	Micro-CT and structured light imaging	Breast	1	No	No	Yes	No	Yes	-	Yes	-	-
Schaaf et al. [28], 2017	Flat-panel volume computed tomography	Head and neck	1	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No
Elastic scattering spectroscopy												
Baykara et al. [29], 2014	Elastic light single scattering spectroscopy	Prostate	2a	No	No	No	No	No	Yes	Yes	No	No
Brouwer de Koning et al. [30], 2018	Diffuse reflectance spectroscopy	Head and neck	2a	No	No	No	No	No	Yes	-	No	No
Brown et al. [31], 2013	Quantitative diffuse reflectance imaging	Breast	2a	No	No	No	No	Yes	Yes	Yes	No	-

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA[2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale[3]	Integration into surgical workflow	Translatable to intraoperative action
De Boer et al.[32], 2015	Diffuse reflectance spectroscopy	Breast	2a	No	No	No	No	-	Yes	Yes	No	No
De Boer et al.[33], 2016	Diffuse reflectance spectroscopy	Breast	2a	No	-	No	No	-	Yes	-	No	No
Grillone et al.[34], 2017	Elastic light single scattering spectroscopy	Head and neck	2a	No	No	No	No	No	Yes	Yes	Yes	No
Lay et al.[35], 2016	Light reflectance spectroscopy	Prostate	2a	No	No	No	No	Yes	Yes	-	No	No
Nichols et al.[36], 2015	Quantitative diffuse reflectance imaging	Breast	2a	No	No	Yes	No	Yes	Yes	Yes	No	-
Fluorescence												
Aoki et al.[37], 2018	Fluorescence (indocyanine green)	Liver Live metastases	2b	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	No
Catapano et al.[38], 2017	Fluorescence (fluorescein sodium)	Brain	2b	Yes	No	Yes	Yes	-	Yes	Yes	Yes	Yes
Inoue et al.[39], 2014	Fluorescence (5-aminolevulinic acid)	Liver Liver metastases	2b	Yes	-	Yes	Yes	No	Yes	Yes	Yes	Yes
Tummers et al.[40], 2014	Fluorescence (methylene blue)	Breast	2b	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Yano et al.[41], 2017	Fluorescence (5-aminolevulinic acid)	Brain	2b	Yes	-	Yes	Yes	-	Yes	Yes	Yes	Yes
Yano et al.[41], 2017	Fluorescence (fluorescein sodium)	Brain	2b	Yes	-	Yes	Yes	-	Yes	Yes	Yes	Yes
Dintzis et al.[42], 2019	Fluorescence (Tozuleristide (BLZ-100))	Breast	2a	No	No	Yes	Yes	Yes	Yes	-	Yes	No
Lamberts et al.[43], 2017	Fluorescence (bevacizumab–IRDye800CW)	Breast	2a	No	No	Yes	Yes	-	Yes	Yes	Yes	Yes
Lee et al.[44], 2017	Fluorescence (indocyanine green)	Brain metastases	2a	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	-
Lieto et al.[45], 2018	Fluorescence (indocyanine green)	Liver Liver metastases	2a	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mondal et al.,[46], 2017	Fluorescence (indocyanine green)	Breast	2a	Yes	-	Yes	Yes	-	Yes	Yes	Yes	No
Neira et al.[47], 2017	Fluorescence (fluorescein sodium)	Brain	2a	Yes	-	Yes	Yes	-	Yes	Yes	Yes	No
Newton et al.[48], 2018	Fluorescence (indocyanine green)	Pancreas	2a	Yes	-	Yes	Yes	-	Yes	Yes	Yes	No

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA[2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale[3]	Integration into surgical workflow	Translatable to intraoperative action
Phipps et al. [49], 2017	Fluorescence lifetime imaging	Breast	2a	No	-	Yes	No	No	Yes	Yes	No	No
Richter et al. [50], 2017	Fluorescence spectroscopy based hand-held probe (5-aminolevulinic acid)	Brain	2a	No	No	No	No	-	Yes	Yes	Yes	No
Richter et al. [50]2017	Fluorescence (5-aminolevulinic acid)	Brain	2a	Yes	-	Yes	Yes	-	Yes	Yes	Yes	Yes
Smith et al. [51], 2018	Fluorescence (LUM015)	Breast	2a	-	No	No	-	-	Yes	-	Yes	No
Yagi et al. [52], 2017	Fluorescence (5-aminolevulinic acid)	Brain metastases	2a	Yes	No	Yes	Yes	-	Yes	Yes	Yes	No
Barabino et al. [53], 2016	Fluorescence (indocyanine green)	Liver metastases	1	Yes	-	Yes	Yes	No	Yes	Yes	No	No
Eyupoglu et al. [54], 2015	Fluorescence (5-aminolevulinic acid, indocyanine green)	Brain	1	No	-	Yes	Yes	-	Yes	Yes	Yes	Yes
Guzzo et al. [55], 2016	Fluorescence (folate-fluorescein isothiocyanate)	Kidney	1	Yes	-	Yes	Yes	-	Yes	Yes	Yes	-
Hekman et al. [56], 2018	Fluorescence (111In-DOTA-girentuximab-IRDye800CW)	Kidney	1	No	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Holt et al. [57], 2015	Fluorescence (indocyanine green)	Soft tissue sarcoma	1	No	-	Yes	Yes	-	Yes	-	Yes	-
Lee et al. [58], 2016	Fluorescence (indocyanine green)	Brain	1	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	-
Li et al. [59], 2018	Fluorescence (68-Ga-IRDye800CW-bombesin)	Brain	1	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predina et al. [60], 2018	Fluorescence (indocyanine green)	Soft tissue sarcoma	1	No	-	Yes	Yes	-	Yes	Yes	Yes	-
Ueo et al. [61], 2015	Fluorescence (gammaglutamyl hydroxymethyl rodamine green)	Breast	1	Yes	-	Yes	Yes	-	Yes	-	No	No
Ujiie et al. [62], 2017	Fluorescence (indocyanine green)	Lung	1	Yes	-	Yes	Yes	-	Yes	Yes	Yes	Yes
Unkart et al. [63], 2017	Fluorescence (AVB-620)	Breast	1	Yes	No	Yes	Yes	-	Yes	Yes	No	No
Van Keulen et al. [64], 2019	Fluorescence (panitumumab-IRDye800CW)	Head and neck	1	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No
Warram et al. [65], 2016	Fluorescence (cetuximab-IRDye800CW)	Head and neck	1	No	-	Yes	No	Yes	Yes	-	No	No
Whitley et al. [66], 2016	Fluorescence (LUM015)	Breast Soft tissue sarcoma	1	-	-	-	-	-	Yes	-	-	-

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA[2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale[3]	Integration into surgical workflow	Translatable to intraoperative action
Hyperspectral imaging												
Fei et al. [67], 2017	Hyperspectral imaging	Head and neck	1	No	-	-	No	-	Yes	-	No	No
Mass spectrometry												
Alfaro et al. [68], 2016	Touch spray mass spectroscopy	Kidney	2a	No	No	No	No	No	Yes	Yes	No	-
Eberlin et al. [69], 2014	Desorption electrospray ionization mass spectroscopy imaging	Pancreas	2a	No	No	-	No	No	Yes	-	No	No
Eberlin et al. [70], 2016	Desorption electrospray ionization mass spectroscopy imaging	Stomach	2a	No	No	-	No	No	Yes	No	No	No
Pirro et al. [71], 2017	Desorption electrospray ionization mass spectroscopy imaging	Brain	2a	No	-	Yes	-	No	Yes	Yes	Yes	No
Balog et al. [72], 2013	Rapid evaporative ionization mass spectrometry	Brain Breast Colorectal Liver Lung Stomach	1	Yes	No	No	No	No	Yes	Yes	Yes	No
MRI												
Roder et al. [73], 2013	MRI	Brain	2b	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Golshan et al. [74], 2014	MRI	Breast	2a	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Heidkamp et al. [75], 2018	MRI	Prostate	2a	No	No	Yes	Yes	Yes	Yes	No	No	No
Heidkamp et al. [76], 2019	MRI	Vulva	2a	No	No	Yes	Yes	Yes	Yes	No	No	No
Mesko et al. [77], 2016	MRI	Soft tissue sarcoma	2a	No	No	Yes	Yes	Yes	Yes	No	Yes	-
Roder et al. [78], 2014	MRI	Brain	2a	No	No	No	No	Yes	Yes	No	Yes	No
Van Oostenbrugge et al. [79], 2019	MRI	Kidney	2a	No	No	Yes	Yes	Yes	Yes	No	No	No
Gamboa et al. [80], 2018	MRI	Brain	1	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Papa et al. [81], 2016	MRI	Breast	1	No	No	Yes	No	Yes	Yes	Yes	Yes	No

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA[2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale[3]	Integration into surgical workflow	Translatable to intraoperative action
Steens et al. [82], 2017	MRI	Head and neck	1	No	No	Yes	Yes	Yes	Yes	No	No	No
Nuclear medicine imaging												
Bluemel et al. [83], 2015	Freehand single photon emission computed tomography (^{99m}Tc-nanocolloid)	Breast	2b	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Grootendorst et al. [84], 2017	Cerenkov luminescence imaging (^{18}F-FDG)	Breast	1	No	No	Yes	No	Yes	Yes	Yes	Yes	No
Watanabe et al. [85], 2018	Specimen positron emission mammography (^{18}F-FDG)	Breast	1	No	No	Yes	Yes	Yes	Yes	Yes	-	-
Optical coherence tomography												
Ha et al. [86], 2018	Optical coherence tomography	Breast	2b	No	No	-	Yes	Yes	Yes	Yes	No	No
Allen et al. [87], 2018	Full field optical coherence tomography	Breast	2a	No	Yes	Yes	No	No	Yes	No	No	No
Assayag et al. [88], 2014	Full field optical coherence tomography	Breast	2a	No	Yes	No	No	No	Yes	Yes	No	No
Chin et al. [89], 2017	Optical coherence elastography	Breast	2a	No	Yes	Yes	No	Yes	Yes	-	-	-
Erickson-Bhatt et al. [90], 2015	Optical coherence tomography	Breast	2a	No	Yes	No	Yes	Yes	Yes	Yes	Yes	No
Hamdoon et al., [91], 2016	Optical coherence tomography	Head and neck	2a	No	Yes	No	Yes	Yes	Yes	-	No	No
Jain et al. [92], 2013	Full field optical coherence tomography	Lung	2a	No	Yes	Yes	No	No	Yes	Yes	No	No
Kennedy et al. [93], 2015	Optical coherence elastography	Breast	2a	No	Yes	Yes	No	No	Yes	Yes	No	No
Ludwig et al. [94], 2018	Optical coherence tomography	Kidney	2a	No	No	No	-	Yes	Yes	Yes	Yes	No
South et al. [95], 2014	Optical coherence tomography	Breast	2a	No	Yes	-	No	-	Yes	Yes	No	-
Van Manen et al. [96], 2017	Full field optical coherence tomography	Pancreas	2a	No	Yes	-	No	Yes	Yes	Yes	No	No
Zysk et al. [97], 2015	Optical coherence tomography	Breast	2a	No	Yes	No	Yes	Yes	Yes	Yes	Yes	No
Sudheendran et al. [98], 2014	Optical coherence tomography	Soft tissue sarcoma	1	No	Yes	-	-	Yes	Yes	-	No	No
pH measurement												
Schartner et al. [99], 2016	Fiber-tip pH probe	Breast Skin	1	No	No	No	No	-	Yes	Yes	No	No

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA [2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale [3]	Integration into surgical workflow	Translatable to intraoperative action
Photoacoustic imaging												
Li et al. [100], 2018a	Photoacoustic tomography	Breast	2a	No	No	Yes	Yes	Yes	Yes	Yes	No	No
Goh et al. [101], 2018	Multispectral optoacoustic tomography	Breast	1	No	No	Yes	Yes	Yes	Yes	Yes	No	No
Radiography												
Hisada et al. [102], 2016	Specimen radiography	Breast	4	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Laws et al. [103], 2018	Specimen radiography	Breast	4	-	-	-	Yes	Yes	Yes	Yes	Yes	-
Majdak-Paredes et al. [104], 2015	Specimen radiography	Breast	4	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Naz et al. [105], 2018	Specimen radiography	Breast	4	No	No	Yes	Yes	Yes	Yes	Yes	-	No
Lange et al. [106], 2016	Specimen radiography	Breast	3	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Amer et al. [107] , 2017	Digital breast tomosynthesis	Breast	2b	No	No	Yes	Yes	Yes	Yes	-	Yes	No
Amer et al. [107], 2017	Full field digital mamography	Breast	2b	No	No	Yes	Yes	Yes	Yes	-	Yes	Yes
Ihrai et al. [108], 2014	Specimen radiography	Breast	2b	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Leung et al. [109], 2015	Specimen radiography	Breast	2b	No	No	Yes	No	Yes	Yes	Yes	Yes	-
Ntomouchtsis et al. [110], 2013	Specimen radiography	Head and neck	2b	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pop et al. [111], 2018	Specimen radiography	Breast	2b	No	No	Yes	Yes	Yes	Yes	Yes	Yes	-
Urano et al. [112], 2016	Digital breast tomosynthesis	Breast	2a	No	No	Yes	Yes	Yes	Yes	-	-	No
Urano et al. [112], 2016	Specimen radiography	Breast	2a	No	No	Yes	Yes	Yes	Yes	-	-	No
Raman spectroscopy												
Barroso et al. [113], 2016	Raman spectroscopy	Head and neck	2b	No	No	No	No	No	Yes	No	No	No
Aubertin et al. [114], 2018	Raman spectroscopy	Prostate	2a	No	No	No	No	-	Yes	Yes	No	No
Barroso et al. [115], 2018	Raman spectroscopy	Head and neck	2a	No	No	Yes	No	No	Yes	No	No	No
Wang et al. [116], 2017	Raman encoded molecular imaging	Breast	2a	No	No	Yes	-	-	Yes	Yes	-	No

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA [2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale [3]	Integration into surgical workflow	Translatable to intraoperative action
Sudheendran et al. [98], 2014	Line scan raman spectroscopy	Soft tissue sarcoma	1	No	Yes	-	-	No	Yes	-	No	No
Terahertz imaging												
Cassar et al. [117], 2018	Terahertz time-domain spectroscopy	Breast	2a	No	No	Yes	No	-	Yes	No	No	No
Grootendorst et al. [118], 2017a	Terahertz pulsed imaging	Breast	2a	No	No	No	No	-	Yes	Yes	Yes	No
Fan et al. [119], 2017	Terahertz pulsed imaging + wide-field polorization enhanced optical imaging	Skin	1	No	Yes	Yes	Yes	No	Yes	No	No	No
Ultrasound												
Laws et al. [103], 2018	Ultrasound	Breast	4	-	-	-	Yes	Yes	Yes	Yes	Yes	-
Sikosek et al. [120], 2014	Ultrasound	Breast	4	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Vieni, et al. [121], 2018	Ultrasound	Breast	4	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Eggemann et al. [122], 2014	Ultrasound	Breast	3	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Eggemann et al. [123], 2016	Ultrasound	Breast	3	No	No	No	Yes	Yes	Yes	-	Yes	Yes
Karanlik et al. [124], 2015	Ultrasound	Breast	3	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Moschetta et al. [125], 2015	Ultrasound	Breast	3	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Ramos et al. [126], 2013	Ultrasound	Breast	3	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Ramos et al. [127], 2014	Ultrasound	Breast	3	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Vispute et al. [128], 2018	Ultrasound	Breast	3	No	No	No	Yes	Yes	Yes	Yes	Yes	No
Abdelshafy et al. [129], 2018	Ultrasound	Breast	2b	No	No	No	Yes	Yes	Yes	Yes	Yes	-
Alharbi et al. [130], 2016	Ultrasound	Kidney	2b	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Eggemann et al. [131], 2014a	Ultrasound	Breast	2b	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Pop et al. [111], 2018	Ultrasound	Breast	2b	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Veeratterapillay et al. [132], 2015	Ultrasound	Kidney	2b	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Scaranelo et al. [133], 2016	Ultrasound	Breast	2a	No	No	No	Yes	Yes	Yes	Yes	Yes	-

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL –stage	Continuous feedback	Resolution near PA[2]	Automatic acquisition	Scanning of entire area of interst	Subsurface information	Non-destructive technique	Within intraoperative timescale[3]	Integration into surgical workflow	Translatable to intraoperative action
Tarabichi et al. [134], 2018	Ultrasound	Head and neck	2a	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No

Supplementary table 4a Diagnostic accuracy on the assessment of samples of tumor versus normal tissue in feasibility studies.

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL stage	N	TP	TN	FP	FN	Sensitivity	Specificity	PPV	NPV
***Advanced* microscopy**												
Brachtel et al. [1], 2016	Spectrally encoded confocal microscopy	Breast	2a	124	68.25	45.57	3.43	6.75	0.91	0.93	0.95	0.87
Chang et al. [3], 2015	Probe-based confocal laser endomicroscopy (acriflavine hydrochloride)	Breast	2a	850	412	380	45	13	0.97	0.89	0.90	0.97
Tao et al. [6], 2014	Non linear microscopy (acridine orange)	Breast	2a	138					95.4	93.3	89.4	97.1
Bio-impedance												
Dotan et al. [21], 2013	Radio frequency spectroscopy	Prostate	2a	10,57	101	7,145	3,30	24	0.80	0.68	0.03	1.00
Elastic* scattering *spectroscopy												
Brouwer de Koning et al. [30], 2018	Diffuse reflectance spectroscopy	Head and neck	2a						0.89	0.82		
Grillone et al. [34], 2017	Elastic light single scattering spectroscopy	Head and neck	2a	176	65.68	70.00	27.00	12.31	0.84	0.71	0.70	0.85
Fluorescence												
Catapano et al. [38], 2017	Fluorescence (fluorescein sodium)	Brain	2b	-					0.85	0.95		
Lee et al. [44], 2017	Fluorescence (indocyanine green)	Brain metastases	2a	39	27	3	8	1	0.96	0.27	0.77	0.75
Neira et al. [47], 2017	Fluorescence (fluorescein sodium)	Brain	2a						0.76	0.75	0.99	
Phipps et al. [49], 2017	Fluorescence lifetime imaging	Breast	2a	1,457	434	991	32	0	1	0.97	0.93	1
Barabino et al. [53], 2016	Fluorescence (indocyanine green)	Liver metastases	1	170	70.29	91.08	7.92	0.71	0.99	0.92	0.90	0.99
Lee et al. [58], 2016	Fluorescence (indocyanine green)	Brain	1	71	50	9	11	1	0.98	0.45	0.82	0.9
Li et al. [59], 2018	Fluorescence (68-Ga-IRDye800CW-bombesin)	Brain	1	42	31	9	0	2	0.94	1	1	0.82
Warram et al. [65], 2016	Fluorescence (cetuximab-IRDye800CW)	Head and neck	1	359.59	137	177.65	31.35	13.59	0.91	0.85	0.81	0.93
Hyperspectral imaging												

Fei et al. [67], 2017	Hyperspectral imaging	Head and neck	1	16					0.92	0.93		
Mass spectrometry												
Alfaro et al. [68], 2016	Touch spray mass spectroscopy	Kidney	2a	29					0.99	0.94		
MRI												
Roder et al. [73], 2013	MRI	Brain	2b	22	17	4	1	0	1	0.8	0.94	1
Papa et al. [81], 2016	MRI	Breast	1	77	31.85	39.06	2.94	3.15	0.91	0.93	0.92	0.93
Optical coherence tomography												
Ha et al. [86], 2018	Optical coherence tomography	Breast	2b	40					0.81	0.89	0.84	0.88
Assayag et al. [88], 2014	Full field optical coherence tomography	Breast	2a	74	28	34	9	3	0.90	0.79	0.76	0.92
Van Manen et al. [96], 2017	Full field optical coherence tomography	Pancreas	2a	98	29	42	15	12	0.71	0.74	0.66	0.78
pH measurement												
Schartner et al. [99], 2016	Fiber-tip pH probe	Breast Skin	1	94					0.88	0.9		
Raman spectroscopy												
Aubertin et al. [114], 2018	Raman spectroscopy	Prostate	2a	477	68	354	39	16	0.81	0.90	0.64	0.96
Barroso et al. [115], 2018	Raman spectroscopy	Head and neck	2a						0.95	0.87		
Wang et al. [116], 2017	Raman encoded molecular imaging	Breast	2a						0.89	0.92		
Terahertz imaging												
Grootendorst et al. [118], 2017a	Terahertz pulsed imaging	Breast	2a	257	99	94	48	16	0.86	0.66	0.67	0.85

Supplementary table 4b Diagnostic accuracy on margin assessment in proof-of-concept studies.

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL stage	Definition of inadequate margin	Missed tumors	N	TP	TN	FP	FN	Sensitivity	Specificity	PPV	NPV
***Advanced* microscopy**														
Espinasse et al. [4], 2017	Fluorescence confocal microscopy	Skin	2a	Positive + close	-	42	0	42	0	0		1		1
Bio-impedance														
Schnabel et al. [18], 2014	Radio frequency spectroscopy	Breast	3	Positive + close	Included						0.75	0.46		
Dixon et al. [19], 2016	Bio-impedance spectroscopy	Breast	2b	Positive + close	Excluded	335	96	170	55	14	0.87	0.76	0.65	0.92
CT														
McClatchy et al. [23], 2018	Micro CT	Breast	2a	Positive	Included	28	2	16	8	2	0.5	0.67	0.2	0.89
Qiu et al. [24], 2018	Micro CT	Breast	2a	Positive + close	Excluded	29	5	20	0	4	0.56	1	1	0.83
Tang et al. [25], 2013	Micro CT	Breast	2a	Positive	Included	25	5	18	1	1	0.83	0.95	0.83	0.95
Schaaf et al. [28], 2017	Flat-panel volume computed tomography	Head and neck	1	Positive	Included	40	3	37	0	0	1	1	1	1
Elastic* scattering *spectroscopy														
Baykara et al. [29], 2014	Elastic light single scattering spectroscopy	Prostate	2a	Positive	Included	45	12	30	1	2	0.86	0.97	0.92	0.94
Brown et al. [31], 2013	Quantitative diffuse reflectance imaging	Breast	2a	Positive + close	Excluded	88	34.04	36.12	5.88	11.96	0.74	0.86	0.85	0.75
Lay et al. [35], 2016	Light reflectance spectroscopy	Prostate	2a	Positive	Included						0.66	0.88	0.66	0.91
Fluorescence														
Van Keulen et al. [64], 2019	Fluorescence (panitumumab-IRDye800CW)	Head and neck	1	Positive + close	Excluded	160					0.95	0.89		
Mass spectrometry														
Balog et al. [72], 2013	Rapid evaporative ionization mass spectrometry	Brain	1	Positive	Included	75					0.95	1		
		Breast				389					0.98	0.98		
		Colorectal				105					0.98	0.94		
		Liver				207					0.99	0.92		
		Lung				22					1	1		
		Stomach				66					1	1		

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL stage	Definition of inadequate margin	Missed tumors	N	TP	TN	FP	FN	Sensitivity	Specificity	PPV	NPV
MRI														
Heidkamp et al. [75], 2018	MRI	Prostate	2a	Positive	Excluded	20	1	14	2	3	0.25	0.88	0.33	0.82
Heidkamp et al. [76], 2019	MRI	Vulva	2a	Positive + close	Included	215	64	83	26	42	0.60	0.76	0.71	0.66
Van Oostenbrugge et al. [79], 2019	MRI	Kidney	2a	Positive	Included	9	1	6	2	0	1	0.75	0.33	1
Optical coherence tomography														
Erickson-Bhatt et al. [90], 2015	Optical coherence tomography	Breast	2a	Positive + close	-						0.92	0.92	0.79	0.97
Hamdoon et al., [91], 2016	Optical coherence tomography	Head and neck	2a	Positive	Included	112	17.93	78.3	11.7	4.07	0.82	0.87	0.61	0.95
Zysk et al. [97], 2015	Optical coherence tomography	Breast	2a	Positive + close	Excluded	43	5	13	22	3	0.63	0.37	0.19	0.81
Photoacoustic imaging														
Li et al. [100], 2018a	Photoacoustic tomography	Breast	2a	Positive + close	Included	20	6.00	10.00	3.00	1.00	0.86	0.77	0.67	0.91
Radiography														
Hisada et al. [102], 2016	Specimen radiography	Breast	4	Positive + close	Excluded	141	6	106	6	23	0.21	0.95	0.5	0.82
Naz et al. [105], 2018	Specimen radiography	Breast	4	Positive + close	Included	52	16	25	6	5	0.76	0.81	0.73	0.83
Amer et al. [107], 2017	Digital breast tomosynthesis	Breast	2b	Positive + close	Excluded	69.5					0.77	0.77	0.89	0.61
Amer et al. [107], 2017	Full field digital mamography	Breast	2b	Positive + close	Excluded	41					0.62	0.97	0.98	0.58
Leung et al. [109], 2015	Specimen radiography	Breast	2b	Positive + close	Included	96	16.00	57.99	21.01	1.00	0.94	0.73	0.43	0.98
Ntomouchtsis et al. [110], 2013	Specimen radiography	Head and neck	2b	Positive + close	Included	15	1	14	0	0	1	1	1	1
Pop et al. [111], 2018	Specimen radiography	Breast	2b	Positive + close	Included	83	4	63	11	5	0.44	0.85	0.27	0.93
Ultrasound														
Eggeman et al. [122], 2014	Ultrasound	Breast	3	Positive + close	Included	198					0.53	0.98		
Karanlik et al. [124], 2015	Ultrasound	Breast	3	Positive + close	Included	84	7	56	17	4	0.64	0.77	0.29	0.93
Moschetta et al. [125], 2015	Ultrasound	Breast	3	Positive + close	Included	132	16	90	6	20	0.44	0.94	0.73	0.82
Ramos et al. [126], 2013	Ultrasound	Breast	3	Positive + close	Included	225	24	116	79	6	0.8	0.59	0.23	0.95
Ramos et al. [127], 2014	Ultrasound	Breast	3	Positive + close	Included	58	9	26	20	3	0.75	0.57	0.31	0.90

Study	Imaging technique (fluorescent/contrast agent)	Cancer type	IDEAL stage	Definition of inadequate margin	Missed tumors	N	TP	TN	FP	FN	Sensitivity	Specificity	PPV	NPV
Alharbi et al. [130], 2016	Ultrasound	Kidney	2b	Positive + close	Excluded	176	3	171	1	1	0.75	0.99	0.75	0.99
Eggeman et al. [131], 2014a	Ultrasound	Breast	2b	Positive + close	Included	882					0.48	0.99	0.81	0.96
Pop et al. [111], 2018	Ultrasound	Breast	2b	Positive + close	Included	83	8	49	25	1	0.89	0.66	0.24	0.98
Scaranelo et al. [133], 2016	Ultrasound	Breast	2a	Positive + close	Included	27	4	13	10	0	1	0.57	0.29	1

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